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Large Language Models (LLMs) for Materials Design

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In recent years, rapid advances in large language models (LLMs) have been witnessed, while materials scientists have quickly adapted to exploit their potential. This review surveys the latest developments at the intersection of LLMs and materials science. Both general-purpose and materials-specific LLMs, as well as their theoretical and technical backgrounds, are first discussed. Their core capabilities are then detailed, including domain-specific question answering for materials design, automated data extraction, semantic-driven material design, synthesis planning, property prediction, and emphasize on inverse materials design as well as integration with materials simulation (e.g., density function theory and crystal structure prediction) workflows. These capabilities are illustrated through case studies in metals/alloys, metal organic frameworks, glasses/ceramics, photovoltaics, catalysis, and batteries, followed by a discussion of advances in retrieval-augmented generation (RAG) and agent systems for materials science. Finally, current challenges and promising future directions of LLMs for materials science are outlined, especially physics-informed transformer (PIT) or materials-informed transformer (MIT) architecture that may effectively bridge the gap between materials science and LLMs (i.e., "LLMs for materials" and "materials for LLMs").

1. Introduction

Artificial intelligence (AI) has revolutionized scientific research by introducing powerful data-driven methodologies.[1–3] An essential branch of AI is machine learning (ML), which employs computational algorithms to extract patterns from data and build predictive models. Commonly used ML techniques range from fundamental algorithms such as linear regression, support vector machines, and decision trees to more complex ensembles like random forests. These are complemented by advanced neural network architectures, including convolutional and long short-term memory networks, as well as optimization techniques such as genetic algorithms,[4,5] collectively expanding the toolbox of AI.[6,7] AI for Science is an emerging subfield of artificial intelligence that applies advanced computational techniques to accelerate scientific discovery.[8,9] Disciplines such as materials science, physics, chemistry and pharmaceutical research increasingly leverage AI to uncover fundamental mechanisms, optimize experimental processes, and accelerate the design of novel materials.[10,11]

On the other hand, natural language processing (NLP) focuses on computational methods for understanding and generating human language, whereas the textual data for NLP is widely available in literature including paper articles, patents and industrial reports. These foundational principles drive advancements in language models and contribute to interdisciplinary applications including computer vision and multimodal systems. In recent years, large language models (LLMs) have become central in AI research,[12,13] particularly following the breakthrough introduction of ChatGPT in 2022. These models are predominantly built on the transformer architecture, which introduces the "attention is all you need" paradigm.[14–16] Beyond transformers, novel architectures such as Mamba have further expanded the field. Prominent LLM examples include ChatGPT, Gemini, LLaMA, Claude, Grok, Mistral, and DeepSeek, each demonstrating remarkable capabilities in natural language understanding and generation in certain aspects but is unfortunately inferior in certain vertical domains.[17–19]

Materials science investigates the fundamental relationships between composition, processing, structure, properties and performance (PSPP),[20,21] which are systematically characterized via specific hardware equipment and software. From the computational science and natural language processing point-of-view, these relationships are often encoded in structured semantic representations, such as standardized terminologies for material classifications and processing ontologies, necessitating precise linguistic formalization for computational analysis. Materials informatics is an interdisciplinary field integrating AI, data science and materials science to accelerate material discovery and optimization.[22] By leveraging machine learning and large-scale data analytics, this domain enables the extraction of meaningful patterns from complex material datasets (data sparsity and data scarcity), facilitating predictive modeling and

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DOI: 10.1002/adfm.202525897

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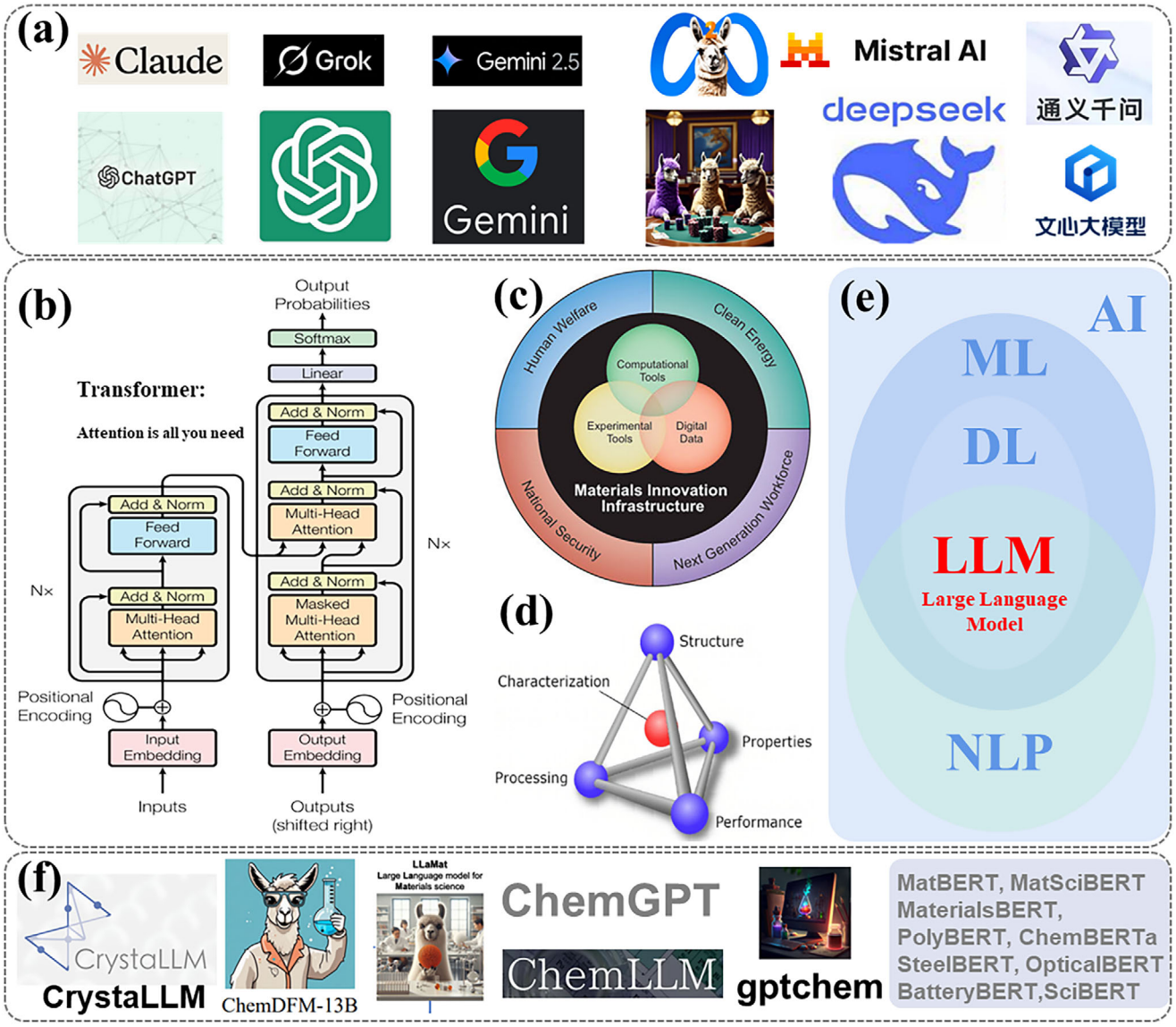


Figure 1. a) Examples of general LLM models. b) Schematic illustration of the Transformer architecture.[37] c) Conceptual framework of MGI. d) Fundamental elements of materials science, highlighting the interrelationships between material composition, structure, processing, properties, performance (PSPP) and characterization. e) Hierarchical relationships between AI, ML, DL, NLP and LLMs. f) Examples of materials-related LLMs.

high-throughput screening. The Materials Genome Initiative (MGI) is a highly influencing initiative that draws inspiration from biological genomics, conceptualizing material systems through analogous "genes" that govern composition-processing-structure-property-performance relationships.[23] Key to this framework is the identification and characterization of these fundamental linkages.[24,25] Large-scale collaborative efforts such as the Materials Project provide extensive computational material databases derived from first-principles calculations based on Schrödinger equation via density-functional theory (DFT), serving as critical resources for data-driven materials research.

Material-specific LLMs are emerging as transformative tools in materials research and development (**Figure 1**).[26] By leveraging domain knowledge and domain-adapted architectures,[27,28] these specialized LLMs enable hypothesis generation and accelerate materials discovery.[29] Progressively, materials researchers are incorporating chatbot-assisted frameworks into computational and experimental workflows to facilitate materials hypothesis generation and data interpretation.[30] Furthermore, the semantic embedding capabilities of LLMs are being harnessed to enhance reasoning and inference in materials informatics applications.[31] Beyond research applications, these models are revolutionizing materials education where scientists and educators are actively integrating LLMs into laboratory instruction and curricula to deliver complex concepts in materials chemistry and crystallography.[32] Materials enterprises are also adopting domain-specific LLMs for materials science to facilitate inverse design, property prediction and various downstream applications. This technological adoption aligns with the core objectives

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of MGI, which aims to halve the development time and associated costs-a strategic imperative that directly addresses materials industry demands for enhanced efficiency and cost reduction.

In this review, we comprehensively examine recent advancements in LLMs for materials design. We systematically organize the discussion into the following key areas: 1) architectural principles of general LLMs and their material-adapted variants; 2) functionalities of materials LLMs including question–answering (Q&A) chatbots, unstructured data extraction, semantic analysis such as LLM-enhanced dimensionality reduction and clustering; and 3) cutting-edge implementations and deployment of LLMs in materials domain such as synthesis planning, crystal structure prediction (CSP), property prediction, and inverse design, with representative case studies across metals/alloys, glasses/ceramics, polymers, photovoltaics, catalysts, batteries, metal-organic frameworks (MOFs), and other critical material domains. We further explore emerging LLM-derived methodologies like retrieval-augmented generation (RAG) and autonomous research agents, while dedicating attention to future directions, such as physics-informed and next-generation transformer architectures for material science. Our focus remains on language-centric large models where semantics are critical, excluding non-linguistic large models (e.g., large atomic potential models).[33–36]

2. Technical Backgrounds

2.1. Basics of LLMs

2.1.1. NLP

The rapid advancement of LLMs is built upon fundamental progress in NLP, which has evolved from early statistical methods to modern neural network representation learning. Initial NLP techniques for scientific studies rely on term frequency–inverse document frequency (TF–IDF) statistics and local context modeling using n-grams. Subsequent advances in neural network-based language models, such as Word2Vec enable word embeddings that capture semantic relationships between words. These developments, in combination with data-driven methods, pave the way for contemporary LLMs, which leverage large-scale pretraining and fine-tuning with billions of parameters fused in the attention-aware transformer architecture to achieve contextual understanding and generation capabilities in domain-specific texts.[38,39]

2.1.2. ML and DL

ML focuses on developing systems that learn from data rather than following pre-programmed rules. Common paradigms include supervised learning (e.g., classification and regression), unsupervised learning (e.g., clustering and dimensionality reduction), and semi-supervised learning. A typical ML workflow encompasses data collection, preprocessing (e.g., handling missing values, normalization, and feature engineering), model training and evaluation using metrics (MAE, RMSE and R^2 for regression; confusion matrices, ROC curves, accuracy, precision, recall for classification). The performance is optimized through loss functions that guide the learning process. DP utilizes multi-layered neural networks to automatically learn hierarchical representations from raw data, enhancing performance in tasks such as natural language processing and image recognition, which are increasingly leveraged in materials science research. The narrative structure of scientific text aligns naturally with autoregression that can predict each token from its preceding context, whereas embedding methods such as Word2Vec learn vector representations of words via context prediction. The hierarchical relationships between AI, ML, DL, NLP, and LLM are presented in Figure 1e.

2.1.3. Transformers

The Transformer architecture is renowned for its self-attention mechanism and serves as the foundational framework for most contemporary LLMs. Its core mechanism relies on self-attention through Query, Key, and Value (QKV) matrices, which dynamically weigh the relevance of each token in the sequence. The model consists of an encoder responsible for understanding and embedding input text into dense representations, and a decoder which generates output sequences autoregressively.[40] These embeddings can capture semantic and domain-specific relationships, making them suitable for scientific applications. For example, the architecture enables effective modeling of long-range dependencies in sequential data, such as chemical formulas, synthesis procedures, or property descriptions,[41] thereby driving progress in tasks like information extraction, concept generation and CSP (depending on the file format and data that are fed into the model). Most recent LLMs (e.g., GPT) use an autoregressive architecture focusing predominantly on the decoder part related to understanding, whereas masked language models like BERT are non-autoregressive and bidirectional, focusing on the encoder part related to generation. Despite of the fact that recent research shifts have been made from BERT models to autoregressive models such as GPT/LLaMA, the bidirectional model still excels in textual understanding. While Transformers underpin widely adopted models like GPT, BERT, and LLaMA, alternatives such as state-space models (e.g., Mamba) have emerged, offering efficient alternative sequence modeling with linear computational complexity. Nevertheless, challenges around computational cost, interpretability, and real-time inference remain active areas of the transformer development.

2.1.4. Pretraining and Fine-Tuning

The development of LLMs typically follows a multi-stage training and tuning pipeline: pre-training on broad corpora to acquire general linguistic and conceptual knowledge, followed by domain-specific fine-tuning to enhance scientific accuracy and relevance (e.g., materials science). While general-purpose LLMs such as LLaMA and ChatGPT demonstrate strong foundational capabilities, they often lack the specialized knowledge required for detailed scientific tasks. Fine-tuning addresses this gap by adapting pre-trained models using curated domain data such as Q&A pairs from scientific papers, transforming the broadly knowledgeable model into a materials-focused expert. Fine-tuning strategies commonly employed are Full Fine-Tuning

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(FFT) that updates all model parameters and demands significant computational resources, and Parameter-Efficient Fine-Tuning (PEFT) methods such as Low-Rank Adaptation (LoRA). In particular, LoRA freezes the original pre-trained weights and introduces trainable low-rank matrices, enabling effective adaptation with minimal parameter updates. In materials science, fine-tuning often relies on high-quality Q&A pairs from materials science papers and databases to modify the model parameters with recognized domain knowledge, improving performance on tasks such as information extraction, synthesis planning, and property prediction.[42]

2.1.5. Prompt Engineering

The deployment of fine-tuned LLMs is frequently accompanied by significant prompt engineering efforts to further improve the model performance. This entails designing precise input instructions that incorporate task context, output specifications and examples to guide the generative process. Techniques such as few-shot learning and reasoning are valuable for improving accuracy in structured data extraction and complex predictive tasks.

2.1.6. Chatbots

The integration of LLMs into conversational agents (chatbots) represents a paradigm shift to benefit daily research. These domain-adapted chatbots serve as intelligent intermediaries, translating natural language into precise queries and synthesized insights. By providing an intuitive interface for complex data and models, they provide access to advanced computational resources and benefit the materials development cycle from an initial informed estimate to real-scenario optimization.[43]

Named Entity Recognition (NER) Named Entity Recognition (NER) plays the central role in data-driven language model study for material science,[44,45] which converts unstructured materials science literature into structured and machine-actionable data format. By identifying and classifying entities such as material compositions (e.g., chemical formula), synthesis recipes, crystal structures, physical/chemical properties and functional performance metrics, NER systems facilitate the large-scale extraction of data essential for materials modeling, property prediction and inverse design. For example, text-mined datasets of solid-state synthesis recipes rely on NER to identify precursors, processing conditions and target compounds.[46] The powerful semantic capabilities of LLMs have made them increasingly popular for NER tasks. They are now often used to complement or even replace traditional rule-based NER systems, primarily because their advanced contextual understanding allows them to disambiguate entities with higher accuracy, generalizing better to unseen data or new domains, and identifying entities based on deep semantic patterns rather than superficial syntactic rules. Specialized language Materials Entity Recognition (MER) language models excel at identifying domain-specific entities, including dopants, synthesis routes (e.g., sol-gel, chemical vapor deposition), structural descriptors (e.g., perovskite, spinel), and property numerical values (e.g., bandgap and ionic conductivity with units).[47,48] The performance of these domain-adapted language models over general language models underscores the importance of specialized training for recognizing nuanced materials science entities.[49] These capabilities directly support downstream applications such as predictive synthesis planning, property-oriented screening and inverse design interconnect composition, synthesis, structure, and properties across the published literature.[50]

Data sources for materials science NER span academic articles, books, patents, material databases, e.g., Materials Project, Inorganic Crystal Structure Database (ICSD), and experimental logs. Key data types include unstructured text (e.g., synthesis descriptions and property paragraphs) and structured representations such as tables and chemical notation strings (e.g., SMILES, SELFIES and InChI).[51] These representations enable semantic parsing of materials entities and support the training of NER systems capable of processing textual materials information.[52] Standardized materials schemas and ontologies provide essential semantic frameworks that define and relate key entities (e.g., materials compositions, processes, properties, structures, and performance). Their adoption ensures consistent representation of extracted data, facilitates interoperability across databases and tools, and improves the accuracy of entity recognition and knowledge graph (KG) construction in automated text mining pipelines. In addition, optical character recognition (OCR) plays a role in extracting structured material data from figures and converting depictions of materials, molecules and processes into machine-readable formats.

2.1.7. Retrieval Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) has emerged as an alternative framework for LLMs in materials science by providing domain-specific responses via dynamically retrieved external knowledge. This approach effectively addresses key limitations of conventional LLMs, such as insufficient domain knowledge and hallucination. This is achieved by integrating large-scale data from scientific papers, domain-specific corpora, and structured databases (e.g., Materials Project).[53,54] By combining semantic retrieval with generative inference, RAG enables LLMs to act as knowledge bases and reasoning systems, mirroring the evidence-based approach of human researchers.

2.1.8. Agents

AI Agents represent an effective shift beyond foundational models, enabling LLMs to function as autonomous research assistants in materials science by integrating with external tools and databases.[55] They can execute complex workflows such as retrieving synthesis conditions, invoking simulation software (such as VASP) for property prediction and structure optimization, and recommending new material compositions and synthesis routes, which dynamically bridge knowledge retrieval and generative decision-making.[42]

3. List of LLMs for Materials Science

3.1. Examples of General LLM Models

LLMs are transformer-based neural networks with billions of parameters trained on extensive text corpora. Figure 1a presents

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representative examples of commonly recognized LLM systems, including ChatGPT, LLaMA, Gemini, Claude, Grok, Mistral, Qwen and Deepseek. The fundamental architecture underlying these models is Transformer (Figure 1b–e).[37] Primarily developed by resource-rich commercial companies for text-generation objective, these models have demonstrated remarkable emergent capabilities across diverse domains. For example, models like GPT-4 encode a vast amount of knowledge from textual datasets, functioning as universal generalists with advanced reasoning and natural language capabilities.[42] The development of LLMs holds the potential to fundamentally reshape research methodologies across material science domains, from composition design to performance optimization and synthesis plan.[56]

3.2. Examples of Materials-Related LLMs

Examples of materials-related LLMs (Figure 1f) include MatBERT,[49] CrystaLLM,[57] LLaMAT,[58] MatSciBERT,[59] SciBERT,[60] MaterialsBERT,[61] PolyBERT,[62] TransPolymer,[63] PolyNC,[64] PolyTAO,[65] SteelBERT,[66] AlloyBERT,[67] OpticalBERT,[68] BatteryBERT,[69] CatBERTa,[70] and CatGPT (**Table 1** and **Figure 2**).[71] There are more emerging materials LLMs[72] and agents,[73] while this study discusses a portion of them. Additional examples are materials chemistry-related LLMs, which are less focused on inorganic materials, including ChemDFM,[74] ChemLLM,[75] ChemGPT,[76] ChemBERTa,[77] AlchemBERT,[78] etc., not to mention more general scientific language models like SciBERT.[60] This review mainly focuses on materials-specific models.

For instance, MatBERT is a widely adopted materials language model focusing on the inorganic materials. It is pretrained on a corpus of 2 million materials science papers and 50 million filtered paragraphs, using Masked Language Modeling (MLM) as the training objective and the AdamW optimizer over 5 epochs, with an initial learning rate of $5e^{-5}$ that decayed linearly.[49] Meanwhile, a materials-specific tokenizer (both cased and uncased situations) is available, despite its limitation in handling non-textual data and being highly specialized in inorganic subdomain terminology. Gupta et al. proposed another materials science model, MatSciBERT,[59] a BERT-based model pretrained on materials science literature, and show that it outperforms general-domain scientific LLMs in tasks such as NER and relation classification. Despite its efficacy in text mining, the model remains limited in processing non-textual data (e.g., crystallographic diagrams) and struggles with jargon outside polymer science, leading to reduced performance in specialized subdomains such as metallurgy and inorganic chemistry. Shetty et al. develop MaterialsBERT, a domain-specific language model trained on 2.4 million materials science abstracts, to address the challenge of extracting material structure-property relationships. Their NLP pipeline extracts approximately 300000 material property records from 130000 polymer abstracts in 60 h, enabling non-trivial insights for applications such as fuel cells and polymer solar cells.[61] Domain-specific training and fine-tuning are essential for achieving high accuracy in materials science applications. However, most current LLMs remain limited in handling multimodal data (e.g., spectra, microscopy). While scaling model size can improve performance (e.g., ChemGPT), it often compromises interpretability. On the industrial front, LLM tools need to be increasingly aligned with practical needs, highlighting a shift toward applications that offer tangible cost and efficiency benefits.

There are several materials LLMs demonstrating strong performance in atomic and crystal structures, which are core aspects of materials science. CrystaLLM demonstrated by Antunes et al.[57] It is a crystal structure generation-related LLM built upon the GPT architecture and crystallographic file data. Trained on an extensive dataset comprising millions of Crystallographic Information File (CIF) entries, it takes composition and space group information as input and outputs structures in CIF format, providing an efficient and automated approach to predicting crystal configurations. LLaMat developed by Mishra et al.[58] is a foundational language model for materials science, with pretraining of LLaMA on a vast corpus of materials literature and crystallographic data. It achieves strong performance in structured information extraction from text and tables, and general material language understanding. A specialized variant LLaMat-CIF, demonstrates capability in generating stable crystal structures with coverage across the periodic table.

In the polymer domain, PolyBERT[62] specializes in mapping polymer sequences to their properties by employing PSMILES strings for composition representation. The model is trained on a synthetically generated dataset comprising 100 million hypothetical polymers, derived from over 13000 existing polymer references. This accelerates property prediction compared to conventional manual fingerprinting techniques. Nevertheless, the model performance is constrained by its reliance on the quality of synthetic training data and exhibits limited generalizability beyond polymer-related applications. Additional LLMs for polymers are available, such as TransPolymer[63] and PolyNC[64] (vide infra).

In the metal and alloy domain, Chaudhari et al. introduce AlloyBERT, a transformer encoder based on RoBERTa/BERT that processes human-readable alloy descriptions (e.g., composition and processing conditions of alloy) to predict mechanical properties of alloys such as elastic modulus and yield strength.[67] Su and Jiang et al. introduce SteelBERT,[66] a language model that integrates textual descriptions of composition and processing parameters to predict mechanical properties of steels using BERT, demonstrating high predictive accuracy for yield strength (R^2 = 89.85%) and tensile strength (R^2 = 88.34%). In a case study involving 15Cr austenitic stainless steel, the model facilitates the design of optimized manufacturing sequences that achieved a yield strength of 960 MPa. Nevertheless, these approaches face challenges such as data scarcity for uncommon metal and alloy systems, as well as limitations in structural design.

For more general science, SciBERT[60] developed by Beltagy et al. is an early pioneer with BERT-based language model pretrained on a scientific corpus comprising 1.14 million papers (3.1B tokens), using full-text content rather than abstracts. It introduces a domain-specific vocabulary (scivocab) optimized for scientific text and offers both cased and uncased versions, alongside baseline models using the original BERT vocabulary for comparison.

In chemistry-related domains, ChemGPT[76] developed by Frey et al. is a generative pre-trained transformer for small molecules, based on 10 million unique molecules with over one billion parameters. Its core innovation lies in establishing neural scaling

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Table 1. Representative examples of materials domain-specific and general LLMs and related resources. Materials-focused models include MatBERT, CrystaLLM, LLaMAT, MatSciBERT, SciBERT, MaterialsBERT, PolyBERT, PolyNC, SteelBERT, AlloyBERT, OpticalBERT, BatteryBERT, CatBERTa, and CatGPT. For comparison, non-materials general models and tools (e.g., ChatGPT) as well as chemistry-oriented and general scientific LLMs are also shown. Each entry provides information on Name, Base Type, Parameters (for several models, exact sizes are not fully disclosed; listed sizes represent available estimates), Key Features, Primary Applications, Research Group, Release Year, Code & Data, and Reference.

Model name	Base type	Parameter	Features	Primary applications	Research group	Release year	Codes & Data	Refs.
MatBERT	BERT	110M	Trained on materials science abstracts. Strong performance in materials-specific NER. Used for materials text mining and knowledge graph construction.	A foundational language model for materials science, fine-tuned on scientific abstracts to help researchers extract information from papers.	Lawrence Berkeley National Laboratory (LBNL)	2021	https://github.com/lbnlp/MatBERT	[49]
MatSciBERT	RoBERTa	110M	RoBERTa-based architecture with a new materials-domain tokenizer; Pre-trained on a large corpus of materials science papers; Strong handling of materials terminology, formulas, and structure descriptors	Materials named-entity recognition (NER); Relation extraction (material–property, process–property); Document classification and knowledge graph construction	Indian Institute of Technology Delhi	2022	https://huggingface.co/m3rg-iitd/matscibert	[59]
MaterialsBERT	BERT	110M	Pre-trained on materials science abstracts and full-text papers; Provides domain-adapted embeddings for materials literature	Text classification in materials science; Information extraction from research papers; Baseline model for downstream materials NLP tasks	Georgia Institute of Technology	2023	https://huggingface.co/pranav-s/MaterialsBERT	[61]
PolyBERT	DeBERTa	139M	Chemical "linguist" for polymers, rapid and accurate property prediction, and able to process polymer SMILES strings as a chemical language.	A domain-specific chemical language model for polymer informatics that treats polymer chemical structures as a form of language.	Georgia Institute of Technology	2023	https://huggingface.co/kuelumbus/polyBERT	[62]
PolyNC	BERT	22M	Polymer-centric natural & chemical language understanding Integration of textual and structural polymer data	Rapid prediction of polymer properties Accelerated polymer design and screening	University of Science and Technology of China	2023	https://huggingface.co/hkqiu/PolyNC	[64]
AlloyBERT	RoBERTa	125M	Leveraging RoBERTa and self-attention mechanisms, achieving superior performance for alloys.	A transformer encoder-based model for predicting alloy properties from textual inputs.	Carnegie Mellon University	2024	https://github.com/cakshat/AlloyBERT	[67]
SteelBERT	BERT	110M	Specialized in steel-related text, knowledge extraction from steel research papers.	A domain-specific model for the steel industry, likely a BERT-based model fine-tuned on a corpus of steel-related literature.	University of Science and Technology Beijing	2025	https://www.gatech.edu/	[66]
LLaMAT	LLaMA	7B, 8B...	Multimodal capability for materials science, able to process both text and CIFs.	A materials research assistant developed by extending the Llama family of models, excelling at materials science NLP and structured data extraction.	Indian Institute of Technology Delhi	2024	https://github.com/M3RG-IITD/llamat	[58]

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Table 1. (Continued)

Model name	Base type	Parameter	Features	Primary applications	Research group	Release year	Codes & Data	Refs.
CrystaLLM	GPT-2	120M	A data-driven model trained on a large dataset to generate crystal structures based on material properties, leveraging LLM capabilities for structured output.	Crystal structure prediction: generates correct CIF syntax and reasonable crystal structures.	Indian Institute of Technology Delhi	2024	https://github.com/mat-llm/CrystaLLM	[57]
MatterGPT	GPT	80M	It uses a simplified line-input crystal-encoding system, which is a string-based crystal representation that satisfies both invertibility and invariances (translational, rotational, and permutational)	A generative pre-trained transformer model for inverse design of inorganic materials	Lingnan University	2023	https://huggingface.co/xiaohang07/MatterGPT	
BatteryBERT	BERT	110M	Expert in battery technology, information extraction from battery research papers.	A BERT model pre-trained on a vast corpus of battery research papers and patents, designed to assist in battery-related text mining.	University of Cambridge	2022	https://doi.org/10.1038/s41467-022-35619-2	[69]
OpticalBERT	BERT	110M	Pre-trained on optical-related text for specialized semantic understanding. Generates contextual embeddings optimized for optical domain classification and regression tasks.	A research model likely focused on extracting and analyzing information from optical science literature.	University of Cambridge	2023	https://huggingface.co/opticalmaterials	[68]
CatBERTa	RoBERTa	110M	CatBERTa leverages the power of Transformer-based language models to process human-interpretable textual inputs and predict energies.	A Transformer-based energy prediction model designed for catalyst property prediction	Carnegie Mellon University	2023	https://github.com/hoon-ock/catberta	[70]
CatGPT	GPT-2	25M	GPT-2-based generative model for catalyst structures Text-conditioned and fine-tunable generation	Automated design of heterogeneous catalysts On-demand generation of slab/adsorbate atom configurations	Sogang University	2024	https://github.com/SeoinBack/CatGPT	[71]
ChemLLM	InternLM2	7B, 13B	LLM dedicated to chemistry, including ChemData, a dataset specifically designed for instruction tuning, and ChemBench, a robust benchmark covering nine essential chemistry tasks	An LLM specialized in chemistry-related tasks. It likely handles chemical reactions, molecular properties, and synthesis planning.	Renmin University of China / Institute for AI Industry Research, Tsinghua University	2024	https://chem-llm.github.io/	[75]
ChemGPT	GPT-Neo model	4.7M, 19M, 1.2B	This model was trained on a subset of molecules from PubChem. You can use this model to generate molecules, but it is mostly intended to be used for investigations of the effects of pre-training and fine-tuning on downstream datasets.	A transformer model for generative molecular modeling, which was pretrained on the PubChem10M dataset	Massachusetts Institute of Technology	2022	https://huggingface.co/ncfrey/ChemGPT-1.2B	[76]

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Table 1. (Continued)

Model name	Base type	Parameter	Features	Primary applications	Research group	Release year	Codes & Data	Refs.
ChemBERTa	RoBERTa	10M, 100M	ChemBERTa scales well with pretraining dataset size, offering competitive downstream performance on MoleculeNet and useful attention-based visualization modalities.	A collection of BERT-like models applied to chemical SMILES data for drug design, chemical modelling, and property prediction.	Stanford University	2020	https://github.com/miservilla/ChemBERTa	[77]
AlchemBERT	BERT	110M	BERT-based chemistry model achieves comparable performance to larger models in material property prediction tasks.	ranging from composition tasks to structure prediction using CIF and natural language descriptions of structures	Beijing Information Science and Technology University	2025	https://github.com/WangYuHang-WYH/AlchemBERT	[78]
ChemDFM	LLaMA	8B, 13B	ChemDFM outperforms the open-sourced LLMs in all the typical tasks of chemistry, and even reaches comparable or higher performances to GPT-4.	A pioneering open-sourced dialogue foundation model for Chemistry and molecular science.	Shanghai Jiao Tong University	2024	https://github.com/OpenDFM/ChemDFM	[74]
SciBERT	BERT	110M	SciBERT is trained on papers from the corpus of semanticscholar.org. Corpus size is 1.14M papers, 3.1B tokens. The full text of the papers are used in training.	A BERT model trained on scientific text.	Allen Institute for Artificial Intelligence	2019	https://github.com/allenai/scibert	[60]
ChatGPT-5	Transformer-based LLM	Not publicly disclosed	Native understanding and generation of images, audio, and video; Higher-level logical reasoning and mathematical skills; Capable of processing much longer context windows.	Expected to be OpenAI's next flagship model, aiming for significant leaps in multimodality and reasoning, moving closer to artificial general intelligence (AGI).	OpenAI		https://openai.com/	
Claude-4	Transformer-based LLM	Not publicly disclosed	Trained using "Constitutional AI" methods for safety, harmlessness, and explainability; Known for its long context window capabilities; Excels at handling complex problems that require multi-step reasoning.	Anticipated to be Anthropic's next-generation model, with a focus on AI safety, ethics, and long-context processing.	Anthropic		https://www.anthropic.com/	
Qwen	Transformer-based LLM	7B,14B,72B, etc.	Multilingual: Supports multiple languages. Multimodal: The Qwen-VL series supports vision. Open Source & Commercial: Provides both open-source models and a commercial API.	An open-source large model series developed by Alibaba, known for its strong multilingual and multimodal capabilities. It is one of the most influential open-source models in China.	Alibaba	2023	https://qwenlm.github.io/	[83]
Ernie Bot	Transformer-based LLM	Not publicly disclosed	Understanding and generation within the Chinese ecosystem; Knowledge-enhanced training: Integrated with a large-scale knowledge graph; Multimodal capabilities: Processes text, image, and audio inputs.	A knowledge-enhanced LLM from Baidu that leverages its search engine and knowledge graph advantages, giving it a significant edge in the Chinese language domain.	Baidu	2021	https://yiyan.baidu.com/	[84]

(*Continued*)

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Table 1. (Continued)

Model name	Base type	Parameter	Features	Primary applications	Research group	Release year	Codes & Data	Refs.
Doubao	Transformer-based LLM	Not publicly disclosed	Cost-effective API pricing; High-concurrency performance; Multimodality and specialized models	ByteDance's versatile and highly competitive model, known for its exceptional cost-performance ratio and high-speed inference.	ByteDance		https://www.doubao.com/	[85]
DeepSeek	Transformer-based LLM with Mixture-of-Experts (MoE) architecture.	127B parameters	Mixture-of-Experts architecture; Multilingual & multimodal support; Cost-effective API pricing; Open-source & API availability	High-efficiency inference; Cross-lingual and vision-language tasks; Budget-conscious commercial deployment; Academic research and development	DeepSeek AI	2025	https://www.deepseek.com/	[86]
Grok	Transformer-based LLM.	Not publicly disclosed.	Real-time data access via X platform; enhanced logical reasoning	Processing and analyzing current events; engaging and entertaining interaction; complex problem-solving and inference	xAI		https://x.ai/	
LLaMA	Transformer-based LLM	7B,13B,33B, 65B	A foundational model for the open-source community, with restrictions on commercial use; Achieved performance comparable to larger, closed models with fewer parameters.	Meta AI's first open-source LLM, which was instrumental in fostering the open-source community's growth.	Meta AI		https://ai.meta.com/blog/	[87]
LLaMA2	Transformer-based LLM	7B,13B, 70B	Commercial-Friendly: fully open for both research and commercial use. Larger Dataset: trained on a significantly larger dataset than its predecessor. Context Length: supports a 4k context length.	The successor to LLaMA, it quickly became the go-to choice for the open-source community due to its permissive license and strong performance.	Meta AI		https://ai.meta.com/llama/get-started/	[88]
LLaMA3	Transformer-based LLM	8B,70B,400B	Longer Context: natively supports an 8k context window. New Tokenizer: uses a new, larger vocabulary tokenizer for more efficient encoding.	The next generation of Llama 2, an open-source models trained on a larger dataset.	Meta AI		https://ai.meta.com/blog/meta-llama-3/	[89]
T5	Encoder-Decoder	60M to 11B	Frames all NLP tasks as "text-to-text" generation, simplifying the model's design. Pre-trained on the massive C4 dataset. Can be used for a wide range of tasks, including summarization, question answering, and translation.	A general-purpose NLP model from Google, with text-to-text paradigm.	Google AI		https://ai.googleblog.com/	[90]
BART	Encoder-Decoder	139M to 406M	Denoising Autoencoder: Trained by corrupting text (e.g., shuffling, deleting) and then reconstructing it, excelling in both understanding and generation. Pre-trained on a large dataset.	A denoising autoencoder for pre-training NLP models from Google, which performs exceptionally well on generative tasks like summarization and translation.	Google AI		https://ai.googleblog.com/	[91]

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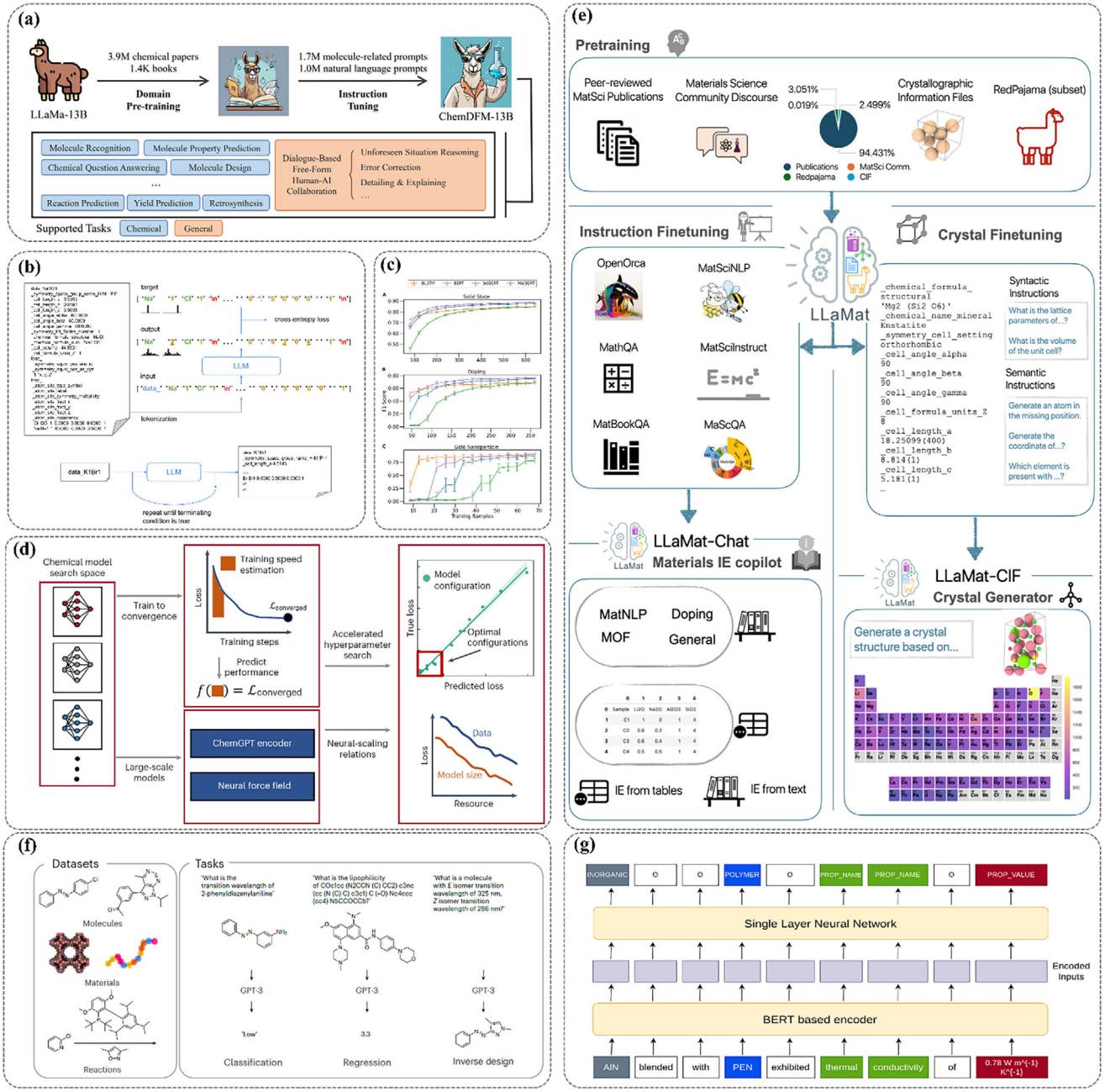


Figure 2. Architectures and features of examples of materials- and materials chemistry-related LLMs. a) ChemDFM, a chemistry-domain foundation model pre-trained on large chemical corpora. Reproduced under the terms of the CC BY 4.0 license. Copyright 2024, Zhao et al.[74] b) CrystaLLM, an LLM based on CIF data (CIF is tokenized into tags, atomic symbols, digits, and punctuation); the model predicts next-token distributions and is optimized by cross-entropy against shifted targets.[61] Reproduced under the terms of the CC BY 4.0 license. Copyright 2023, Shetty et al. c) MatBERT, a BERT-style encoder for materials texts; example shows NER fine-tuning learning curves on solid-state, doping and Au-nanoparticle datasets as training size increases.[49] Reproduced under the terms of the CC BY 4.0 license. Copyright 2022, Trewartha et al. d) ChemGPT, GPT-family model for chemical text and strings, enabling efficient discovery of neural-scaling relations across model and data sizes.[76] Adapted with permission from Ref. [76]. Copyright 2023, Springer Nature. e) LLaMAT, a LLaMA-based model adapted to materials corpora via domain fine-tuning, compatible with retrieval-augmented workflows for literature Q&A and synthesis planning.[58] Reproduced under the terms of the CC BY 4.0 license from Ref. [58]. Copyright 2025, Mishra et al. f) GPTChem, a GPT-style chemistry model leveraging domain prompts and tools for property prediction, reaction reasoning and text-to-structure tasks.[80] Adapted with permission from Ref. [80]. Copyright 2024, Springer Nature Limited. g) MaterialsBERT, a BERT encoder with a token-classification head for materials NER.[61] Reproduced under the terms of the CC BY 4.0 license from Ref. [61]. Copyright 2023, Shetty et al.

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laws across chemical language models and graph neural networks (GNNs) for interatomic potentials. Another similar specialized chemical language model is ChemLLM, which targets molecular understanding and generation tasks.[75] The chemical multimodal LLM ChemMLLM[79] integrates text, SMILES strings, and molecular images to bridge cross-modal understanding gaps. GPTChem[80] proposed by Jablonka et al. is adapted from GPT-3 and demonstrates competitive performance in several regimes (e.g., reaction yield prediction) and enables inverse design via natural language questions. ChemDFM (Figure 1) developed by Zhao et al. represents another open-source dialogue foundation model for chemistry and molecular sciences. This model is based on fine-tuning the LLaMA-13B architecture and achieves competitive and enhanced results relative to GPT-4 in specialized molecular science applications.[74,81] A wide range of other chemistry- and molecule-related LLMs are available, such as MolGPT, BioGPT, AutoMolDesigner, CMolGPT, ChemFormer, MOGEN, BioMistral, Mixtral, ChXagent, Tx-LLM, PaLM-2ReactionT5, BioT5, Text+Chem T5, ChemBERTa, CatBERTa, ChemBERTa-2, SMILES-BERT, PubBedBERT, MolBERT, ScholarBERT, MolBERT, ClinicalBERT, PMC-LLaMa, Darwin, BioMedGPT-LM, ChatDoctor, Tx-LLM, BioMedLM, LlasMol, BioMistral, BiMediX, EpilepsyLLM, CheXagent, ChemSpaceAL, BioMedGPT-LM, Darwin, cMolGPT, PMC-LLaMA, Galactica, BioGPT.[82] On the other hand, LLMs applied to the materials domain[56] exhibit notable limitations, including residues of hallucination, low sensitivity to design context and a tendency to overestimate material suitability. While prompt engineering can improve performance, domain-specific LLM recommendations still lag significantly behind expert decisions. These gaps highlight the continuous need for better domain-specific adaptation to enhance their practical utility in materials design.[27]

4. Functionalities of Materials LLMs

Materials LLMs are evolving from general chat interfaces into research copilots that can answer difficult domain questions and extract structured data (i.e., compositions, processing conditions, structures, and device metrics) from unstructured papers.[92–94] They also help map semantic relationships across materials knowledge by embedding texts and entities in vector spaces that support cosine-similarity search, dimensionality reduction, and clustering, thereby enabling the discovery of latent composition-application or structure-property relationships. Beyond literature mining, materials LLMs assist in synthesis reasoning, forward property prediction[95] and inverse design via targeted sequence generation over materials encodings.[96] These models can be coupled with physics and materials structure data, such as LLM+DFT pipelines that validate proposals and refine predictions, and LLM+CSP approaches that use text-to-CIF generation to initiate symmetry-aware structure search. Finally, their accuracy can be improved by RAG and agent workflows that ground outputs in literature and external tools.[97–99]

4.1. Q&A Chatbots for Materials Design

Question answering (Q&A) chatbots represent one of the most direct applications of LLMs in materials science, offering intuitive access to complex material knowledge and supporting data-driven design decisions. Zheng and Yaghi et al. develop a ChatGPT-based assistant for text mining and predicting the synthesis of metal-organic frameworks (MOFs), incorporating structured prompt engineering (ChemPrompt) and minimizing hallucinations via extracting synthesis parameters (MOF synthesis procedures and reaction conditions) from literature sources (**Figure 3**). This end-to-end workflow spanning retrieval, parsing, tabulation, and dialog, demonstrates how LLM-driven Q&A systems can transform unstructured text into actionable synthesis insights.[100] Regarding the materials LLM chatbots, further efforts are needed to ensure the consistency of Q&A outputs, mitigating the risk of latent errors, and thoroughly evaluating ethical consequences.[93]

4.2. Data Extraction

Automated data extraction from unstructured scientific text (**Figure 4**) is a key application of LLMs in materials science, enabling the construction of structured knowledge essential for subsequent design and analysis.[102] While rule-based NER tools such as ChemDataExtractor are available to extract useful information from materials literature, LLMs offer greater flexibility and tailored performance to obtain structured knowledge, though they may suffer from hallucination at some point. Recent approaches[103,104] leverage both fine-tuning and advanced prompt engineering to accurately parse complex chemical descriptions, not only chemical formulas but also properties and synthesis conditions. For instance, Zhang et al.[105] demonstrated that fine-tuned LLMs excel across diverse tasks, including compound recognition, reaction labeling and MOF synthesis extraction, achieving exact accuracies between 69% and 95% with minimal annotated data. Similarly, conversational strategies such as ChatExtract combine iterative prompting and uncertainty-driven verification to achieve precision and recall approaching 90% in extracting materials properties, effectively mitigating LLM hallucinations.[106]

4.3. Semantic Analysis (Cosine Similarity, Dimensional Reduction, and Clustering)

Embedding-based language models in materials science encode entities such as compositions, processes, and properties into high-dimensional vectors, which are not directly comprehensible by the human mind (e.g., 100–200 dimensions in word2vec and thousands more in modern LLMs). Cosine similarity provides a natural metric to quantify semantic similarity relationships between materials entities based on LLM embeddings (**Figure 5**), enabling fast retrieval of similar materials, properties or applications, and providing a route to predict research trends. Early studies show that unsupervised embeddings in language models capture chemically meaningful relationships and anticipate future discoveries through cosine-based nearest-neighbor search.[108]

Apart from cosine similarity calculations, a wide range of dimensionality reduction methods is widely employed for materials-related understanding and generation. Linear methods such as PCA offer efficient preprocessing, while nonlinear approaches such as t-SNE and UMAP provide 2- or 3-D

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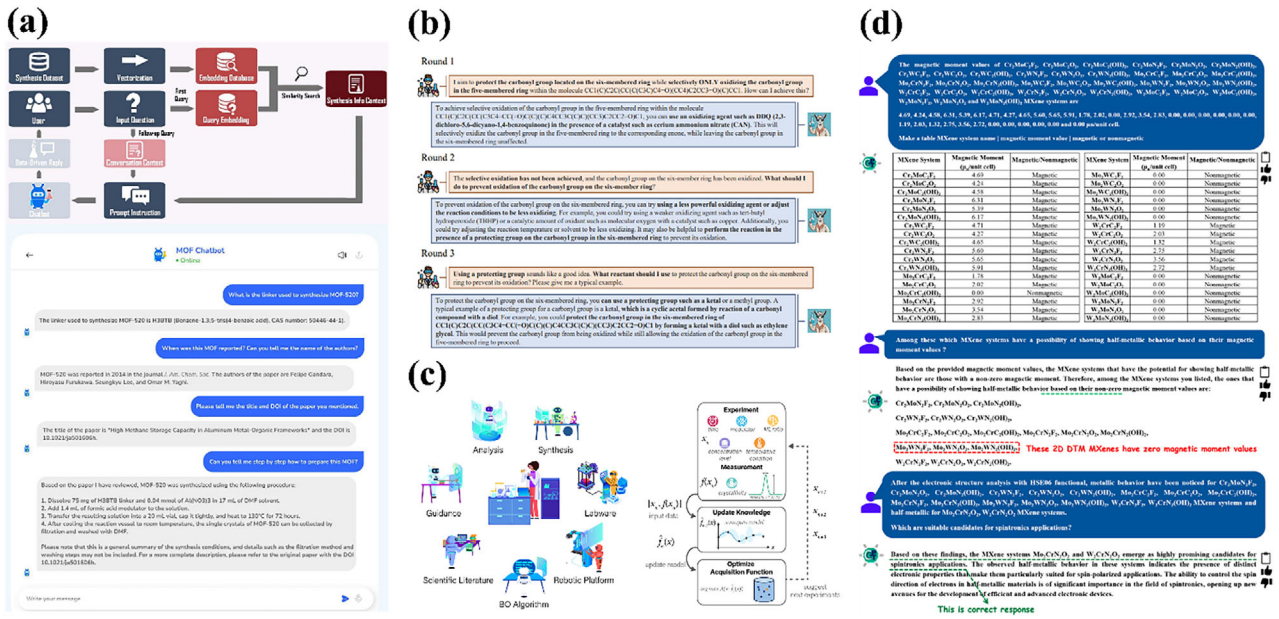


Figure 3. Q&A exemplars with materials-focused LLMs. a) Integrated MOF design workflow that converts comprehensive MOF synthesis datasets into a context-aware dialogue system. Reproduced with permission.[100] Copyright 2023, American Chemical Society. b) Domain-tuned foundation model answering materials chemistry queries; exemplar interactions show literature-grounded guidance for design and synthesis. Reproduced under the terms of the CC BY 4.0 license. Copyright 2024, Zhao et al.[74] c) ChatGPT-based assistants collaborating across stages of synthesis optimization (propose experiments, execute and analyze results, update surrogate model with new data, and optimize acquisition function for next-batch recommendations). Reproduced with permission.[101] Copyright 2023, American Chemical Society. d) Stepwise Q&A workflow for MXene material design. Reproduced with permission.[93] Copyright 2024, American Chemical Society.

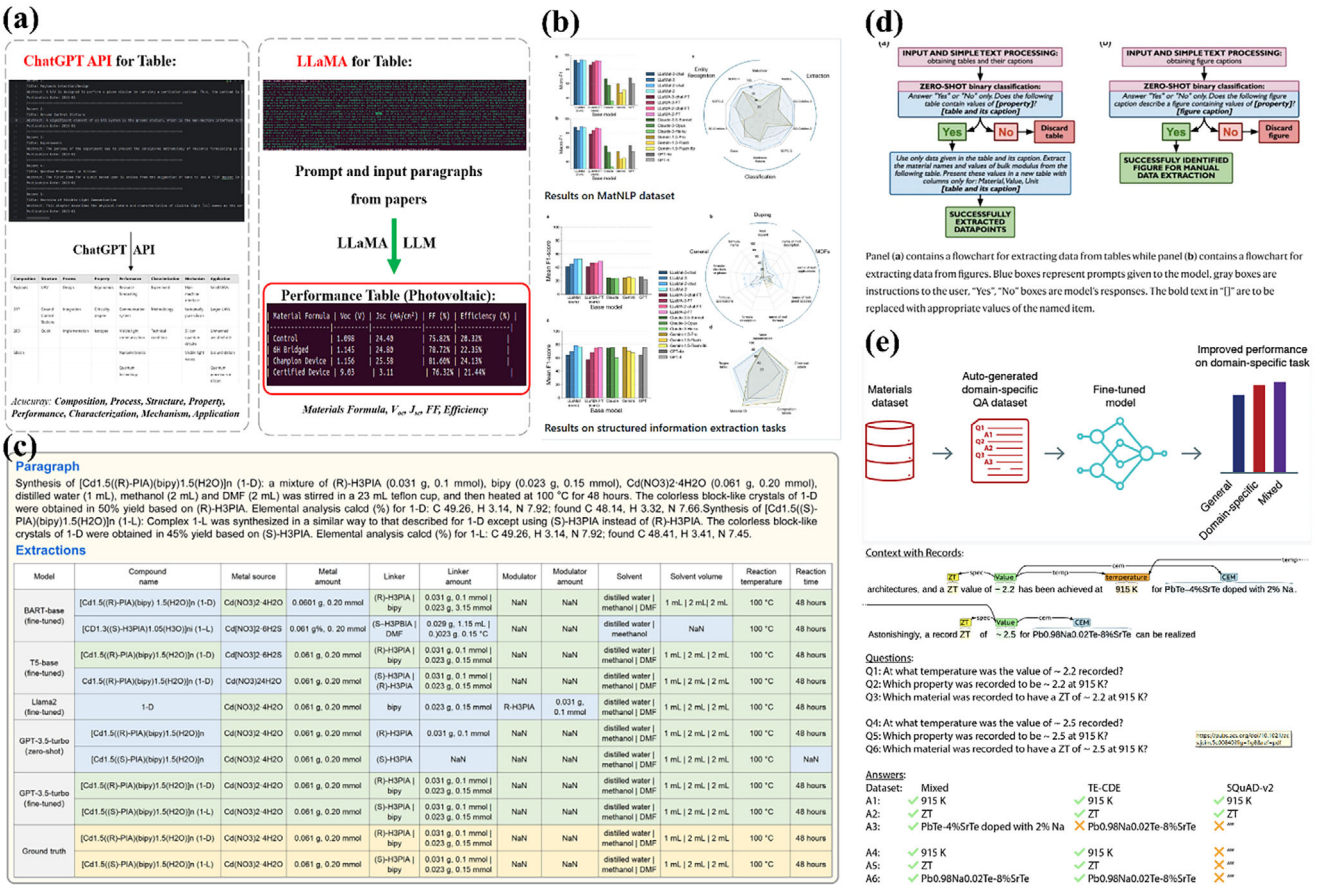
maps that highlight both local and global relationships. These visualizations allow researchers to recognize clusters of materials families or functional domains, bridging machine-learned embeddings with human insight.[109] Clustering (e.g., K-means and DBSCAN/HDBSCAN) further transforms continuous embedding spaces into discrete groups, useful for defining topical domains (e.g., thermoelectrics, perovskites, photovoltaics) or application-driven categories.[109,110] Recent applications demonstrate that embedding-based clusters align with experimentally validated families, making them valuable for candidate screening, such as improving the photovoltaic device performance from 23% to 25%.[111] Zhang et al. employ semantic cosine similarity derived from language model embeddings of the literature as input features to enhance genetic model predictions for processing solvent design; the approach has been experimentally validated, demonstrating improved perovskite film optoelectronic stability based on a language model.[110] As the literature grows and task-specific tuning improves, improved language features may be formulized via hybridizing semantic descriptors and DFT-derived descriptors,[112] while the language-native predictors are likely to become standard components of high-fidelity and physics-informed property models.

4.4. Material Synthesis

Materials synthesis is the most fundamental aspect of materials science and engineering, linking composition design to practical realization of structures and properties for both academic research and industrial applications.[114,115] Recent works highlight the promise of natural language processing or LLMs in addressing the long-standing challenge of representing and predicting complex materials synthesis workflows. For example, a synthesis protocol is proposed that involves not only the selection of material precursors but also a structured sequence of actions such as add, mix, stir, grind, heat, anneal, quench, or cool.[46,48,116] These actions are often coupled with explicit reaction equations that specify stoichiometry and target phases, as well as practical constraints such as heating rates, vacuum/atmosphere control, or doping levels.[117–119] Beyond these operations, synthesis relies on a wide range of processing tools and preparation instruments, including chemical vapor deposition (CVD), sol-gel methods, solid-state reactions, hydrothermal synthesis, thin-film deposition systems, and high-pressure furnaces.[120] Once prepared following structured sequence of actions and fabrication equipment, the resulting materials must be closely examined using characterization instruments such as X-ray diffraction (XRD) for phase identification, scanning electron microscopy (SEM) and transmission electron microscopy (TEM) for microstructural analysis, Raman and infrared spectroscopy for bonding information, or X-ray photoelectron spectroscopy (XPS) for surface chemistry. The combination of precursors, action verbs, reaction equations, synthesis tools, and characterization techniques defines the comprehensive semantics of materials processing. Yet this information is typically scattered across unstructured literature and written in highly heterogeneous formats; LLMs create new opportunities to automatically extract and standardize these elements, represent them as structured workflows, and transform narrative synthesis recipes into flowchart-like process maps.[121–125]

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Model	Compound name	Metal source	Metal amount	Linker	Linker amount	Modulator	Modulator amount	Solvent	Solvent volume	Reaction temperature	Reaction time
BART-base (fine-tuned)	[Cd1.5((R)-PIA)(bipy) 1.5(H2O)]n (1-D)	Cd(NO3)2·4H2O	0.0601 g, 0.20 mmol	(R)-H3PIA | bipy	0.031 g, 0.1 mmol | 0.023 g, 3.15 mmol	NaN	NaN	distilled water | methanol | DMF	1 mL | 2 mL | 2 mL	100 °C	48 hours
	[CD1.3((S)-H3PIA)1.05(H3O)]ni (1–L)	Cd[NO3]2·6H2S	0.061 g%, 0. 20 mmol	(S–H3PBIA | DMF	0.029 g, 1.15 mL | 0.)023 g, 0.15 °C	NaN	NaN	distilled water | meethanol	NaN	100 °C	48 hours
T5-base (fine-tuned)	[Cd1.5((R)-PIA)(bipy)1.5(H2O)]n (1-D)	Cd[NO3]2·6H2S	0.061 g, 0.20 mmol	(R)-H3PIA | bipy	0.031 g, 0.1 mmol | 0.023 g, 0.15 mmol	NaN	NaN	distilled water | methanol | DMF	1 mL | 2 mL | 2 mL	100 °C	48 hours
	Cd1.5((R)-PIA)(bipy)1.5(H2O)]n (1-D)	Cd(NO3)24H2O	0.061 g, 0.20 mmol	(S)-H3PIA | (R)-H3PIA	0.031 g, 0.1 mmol | 0.023 g, 0.15 mmol	NaN	NaN	distilled water | methanol | DMF	1 mL | 2 mL | 2 mL	100 °C	48 hours
Llama2 (fine-tuned)	1-D	Cd(NO3)2·4H2O	0.061 g, 0.20 mmol	bipy	0.023 g, 0.15 mmol	R-H3PIA	0.031 g, 0.1 mmol	distilled water | methanol | DMF	1 mL | 2 mL | 2 mL	100 °C	48 hours
GPT-3.5-turbo (zero-shot)	[Cd1.5((R)-PIA)(bipy)1.5(H2O)]n	Cd(NO3)2·4H2O	0.061 g, 0.20 mmol	(R)-H3PIA	0.031 g, 0.1 mmol	NaN	NaN	distilled water | methanol | DMF	1 mL | 2 mL | 2 mL	100 °C	48 hours
	[Cd1.5((S)-PIA)(bipy)1.5(H2O)]n	Cd(NO3)2·4H2O	0.061 g, 0.20 mmol	(S)-H3PIA	NaN	NaN	NaN	distilled water | methanol | DMF	1 mL | 2 mL | 2 mL	100 °C	NaN
GPT-3.5-turbo (fine-tuned)	[Cd1.5((R)-PIA)(bipy)1.5(H2O)]n (1-D)	Cd(NO3)2·4H2O	0.061 g, 0.20 mmol	(R)-H3PIA | bipy	0.031 g, 0.1 mmol | 0.023 g, 0.15 mmol	NaN	NaN	distilled water | methanol | DMF	1 mL | 2 mL | 2 mL	100 °C	48 hours
	[Cd1.5((S)-PIA)(bipy)1.5(H2O)]n (1-L)	Cd(NO3)2·4H2O	0.061 g, 0.20 mmol	(S)-H3PIA | bipy	0.031 g, 0.1 mmol | 0.023 g, 0.15 mmol	NaN	NaN	distilled water | methanol | DMF	1 mL | 2 mL | 2 mL	100 °C	48 hours
Ground truth	[Cd1.5((R)-PIA)(bipy)1.5(H2O)]n (1-D)	Cd(NO3)2·4H2O	0.061 g, 0.20 mmol	(R)-H3PIA | bipy	0.031 g, 0.1 mmol | 0.023 g, 0.15 mmol	NaN	NaN	distilled water | methanol | DMF	1 mL | 2 mL | 2 mL	100 °C	48 hours
	[Cd1.5((S)-PIA)(bipy)1.5(H2O)]n (1-L)	Cd(NO3)2·4H2O	0.061 g, 0.20 mmol	(S)-H3PIA | bipy	0.031 g, 0.1 mmol | 0.023 g, 0.15 mmol	NaN	NaN	distilled water | methanol | DMF	1 mL | 2 mL | 2 mL	100 °C	48 hours

Figure 4. Structured data extraction with LLMs for materials science. a) In-house LLaMA- and ChatGPT-based extraction from scientific papers, with examples of turning materials text into schema-aligned tables. b) LLaMAT extraction performance on benchmark tasks.[58] Reproduced under the terms of the CC BY 4.0 license from Ref. [58]. Copyright 2025, Mishra et al. c) LLM-based extraction of multi-reaction MOF synthesis paragraph.[105] Adapted under the terms of the CC BY 4.0 license from Ref. [105]. Copyright 2024, Zhang et al. d) Flowcharts for LLM-driven extraction from sentences/paragraphs and tables/figures using ChatExtract.[106] Adapted under the terms of the CC BY 4.0 license from Ref. [106]. Copyright 2024, Polak et al. e) Workflow auto-generating a domain-specific QA dataset from ≈60k articles with compound and property (ZT, Seebeck coefficient, thermal/electrical conductivity, power factor) for thermoelectrics with BERT models.[107]Adapted with permission from Ref. [107]. Copyright 2025, American Chemical Society.

Recent advances show that LLMs can be fine-tuned to predict the synthesizability of inorganic compounds and polymorphs, identify suitable precursors, and propose feasible synthesis details (**Figure 6**).[126,127] For example, Kim and Jung et al.[126] demonstrate that LLMs trained on text descriptions of crystal structures achieve improved accuracy than graph neural networks while offering human-readable explanations of the governing rules to support explainable synthesizability prediction of inorganic crystal polymorphs.[128] Thway et al. demonstrate that GPT-3.5 can be harnessed to parse synthesis recipes of ternary chalcogenides.[127] Enhanced RAG using integrated term enhancement methodology (ITEM) is proposed to extract and explain critical domain terms in materials synthesis research.[129] Text mining with LLMs has been extended to unify heterogeneous synthesis data and provide coding-free interfaces for experimental materials scientists.[100]

Despite these advances, there is a huge gap between theoretical material stability calculation and real-scenario material synthesizability. Several studies have been carried out to show that LLMs can help bridge the gap between experimental synthesizability of materials and the thermodynamic or kinetic stability criteria.[130] Wang et al.[131] introduce the Crystal Synthesis LLM (CSLLM) framework, which integrates fine-tuned LLMs to separately predict the synthesizability of 3D crystal structure, identify plausible synthetic methods, and suggest suitable solid-state precursors. With accuracies up to 98.6%, CSLLM outperformed traditional thermodynamic- or kinetic-based screening and demonstrated strong generalization to experimentally complex structures of materials. In addition, there is a lack of studies to address LLMs' ability to reason about synthesis mechanisms. To address this, Pu et al. propose a benchmark for probing LLMs to explain mechanistic reasoning in nanoparticle synthesis and suggest that RAG improves performance in explaining synthesis mechanisms.[132] Moreover, Zhang et al. combine semantic information derived from language models with DFT results to construct hybrid descriptors capturing chemical and structural mechanisms for material processing, enabling a generative and explainable machine learning framework for materials prediction.[112,133]

Together, these studies underscore the versatility of LLMs in materials synthesis, covering text parsing, bias detection,

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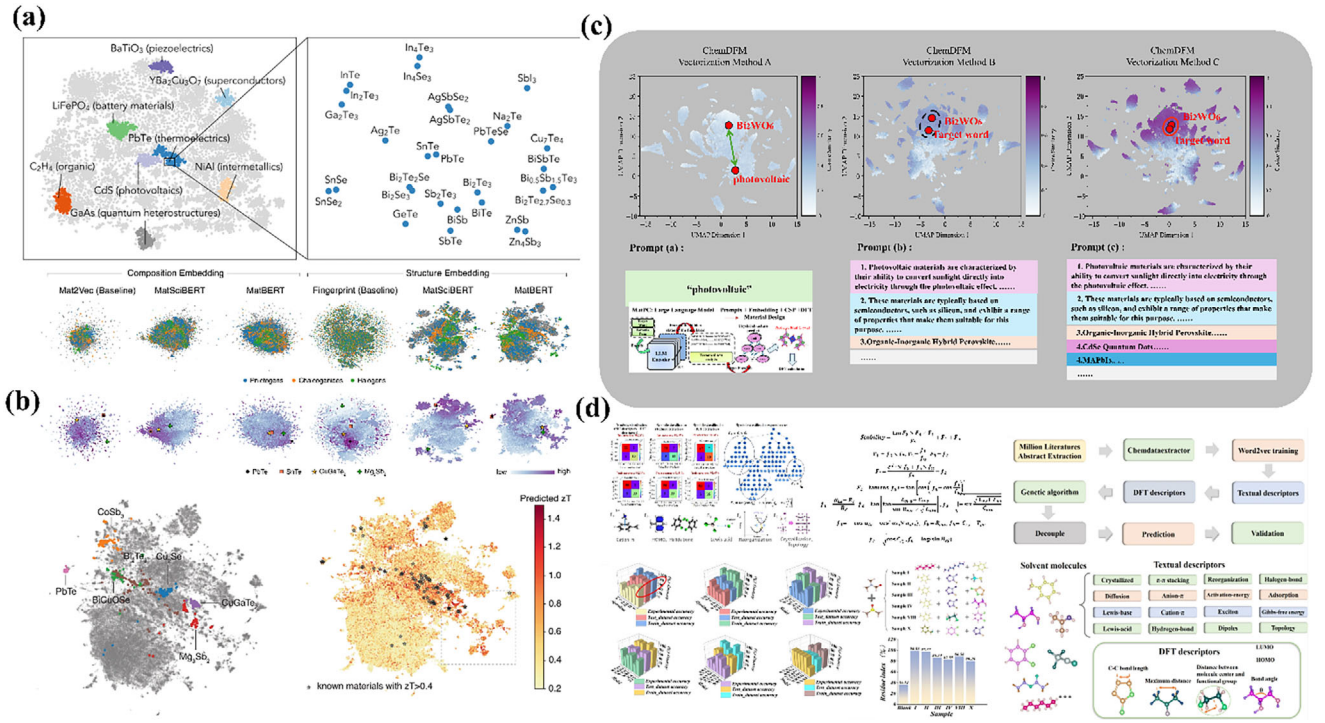


Figure 5. Semantic maps using language models from dimensionality reduction and clustering for materials analysis and fast prediction. a) Dimensionality reduction (e.g., PCA and t-SNE) with cluster structure revealing LLM-based materials embedding relationships.[108] Adapted with permission from Ref. [108]. Copyright 2019, Springer Nature Limited. b) UMAP visualizations built from text embeddings (e.g., sentence and paragraph vectors from domain language models) enable semantic neighborhoods of materials, synthesis routes and properties.[109] Adapted under the terms of the CC BY 4.0 license from Ref. [109]. Copyright 2024, Qu et al. c) MatPC pipeline with an example of coupling LLM-guided semantic screening with crystal structure prediction and DFT analysis, where the language model cluster views aid "smart guess" of candidate generation.[113] Adapted with permission from Ref. [113]. Copyright 2025, American Chemical Society. d) Descriptor design based on language model and DFT calculations for experimental material design. These language data-driven descriptors are integrated into a genetic model to describe material stabilities.[112] Adapted with permission from Ref. [112]. Copyright 2024, American Chemical Society.

synthesizability prediction, precursor suggestion and mechanism explanation. While chemistry-focused models (e.g., retrosynthesis tools such as Chemma[134] and SynAsk[43]) emphasize reaction planning involving molecules,[135] materials synthesis typically consists of a multi-stage flow process involving experimental actions (add, mix, grind, heat), materials precursors and specialized instruments (e.g., high-temperature furnaces, CVD systems, XRD, SEM/TEM). This procedural complexity makes materials synthesis both challenging and rewarding as a target for LLM assistance. Regrettably, most reported synthesis data are biased toward successful experiments, while negative results are rarely available. Despite current limitations such as non-standardized reporting styles and scarcity of negative data, LLMs are beginning to bridge literature-scale knowledge with experimental practice, paving the way toward automated, interpretable and scalable materials synthesis design.

4.5. Property Prediction

LLMs are beginning to act as forward predictors of materials properties, such as mapping textual inputs to quantitative numbers associated with band gaps, formation energy, elastic constants, mechanical properties, efficiencies and capacity. First, LLMs can be fine-tuned to perform direct regression from simple strings (e.g., formulas, SMILES) to target properties. Second, they can read the literature (extracting compositions, measurement conditions and reported values) and perform NER, so that conventional ML or physics-aware models can be employed to predict virtual property space.[136]

Related efforts in alloys and polymers show that transformer encoders trained on domain text can forecast mechanical properties from composition-level inputs, improving the practicality of language-first predictors when detailed structures are unavailable.[67] Another example of direct regression fine-tunes LLaMA-3 on composition/SMILES, achieving decent accuracy on QM9 molecular targets and a suite of materials properties.[137] This reflects the core trade-off between composition-only inputs and structural features from crystal graphs or 3D representations. Domain-tuned property specialists such as ElaTBot is an LLM tailored for elasticity that predicts elastic-constant tensors, finite-temperature bulk moduli and proposes candidates with targeted elastic responses.[138]

Equally important is the role of LLMs in dataset-construction for property prediction. Earlier work on literature embeddings demonstrates that materials texts encode rich structure-property information, while extensive recent materials science studies push this further by using LLMs to mine compositions, units,

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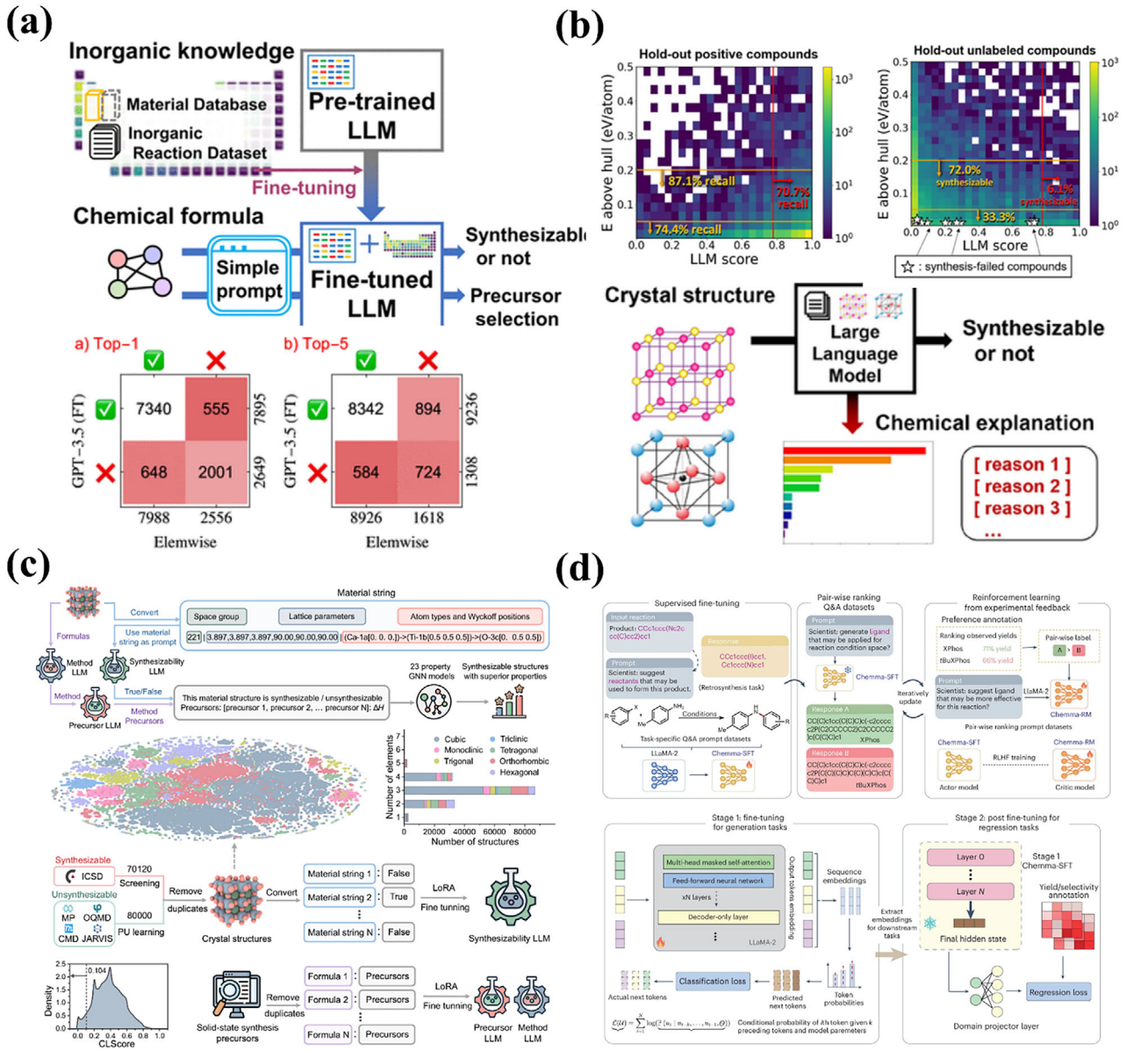


Figure 6. LLMs for materials synthesis. a) Inorganic precursor prediction with fine-tuned GPT.[128] Adapted with permission from Ref. [128]. Copyright 2024, American Chemical Society. b) Explainable synthesizability prediction of material crystal polymorphs using LLMs.[126] Adapted with permission from Ref. [126]. Copyright 2025, Wiley. c) An end-to-end crystal synthesis LLMs framework for predicting synthesizability and recommending precursors and fabrication methods.[131]Adapted under the terms of the CC BY 4.0 license from Ref. [131]. Copyright 2025, Song et al. d) Model architecture and training strategy underpinning Chemma, an LLM fine-tuned for synthesis planning using 1.28 million reaction Q&A pairs.[134]Adapted with permission from Ref. [134]. Copyright 2025, Springer Nature Limited.

protocols, and property values from papers, patents and reports. The resulting tables feed downstream regressors (e.g., linear regression, tree methods, GNNs) for property prediction.[139] MolecularGPT[140] shows few-shot molecular property prediction across hundreds of tasks. Leveraging LLMs for predictive chemistry finds that lightly fine-tuned GPT-style models can outperform specialized ML in a low-data regime and support simple inverse design.[80] Cross-modal LLM+GNN frameworks further fuse language information with graph structure to improve model robustness.[141]

In materials science, many properties depend on periodic crystal structure, symmetry, defects, and processing history. While composition-only text can be informative, high-fidelity models typically require hybridizing LLMs with crystal graphs, DFT features and detailed synthesis conditions. As a result, composition-only LLM regression (e.g., fine-tuned LLaMA-3) provides a usable baseline when structures are missing, whereas domain-tuned models can improve accuracy further when combined with RAG and physics-aware inputs. A practical workflow is to use LLMs for few-shot generalization and dataset curation, and then

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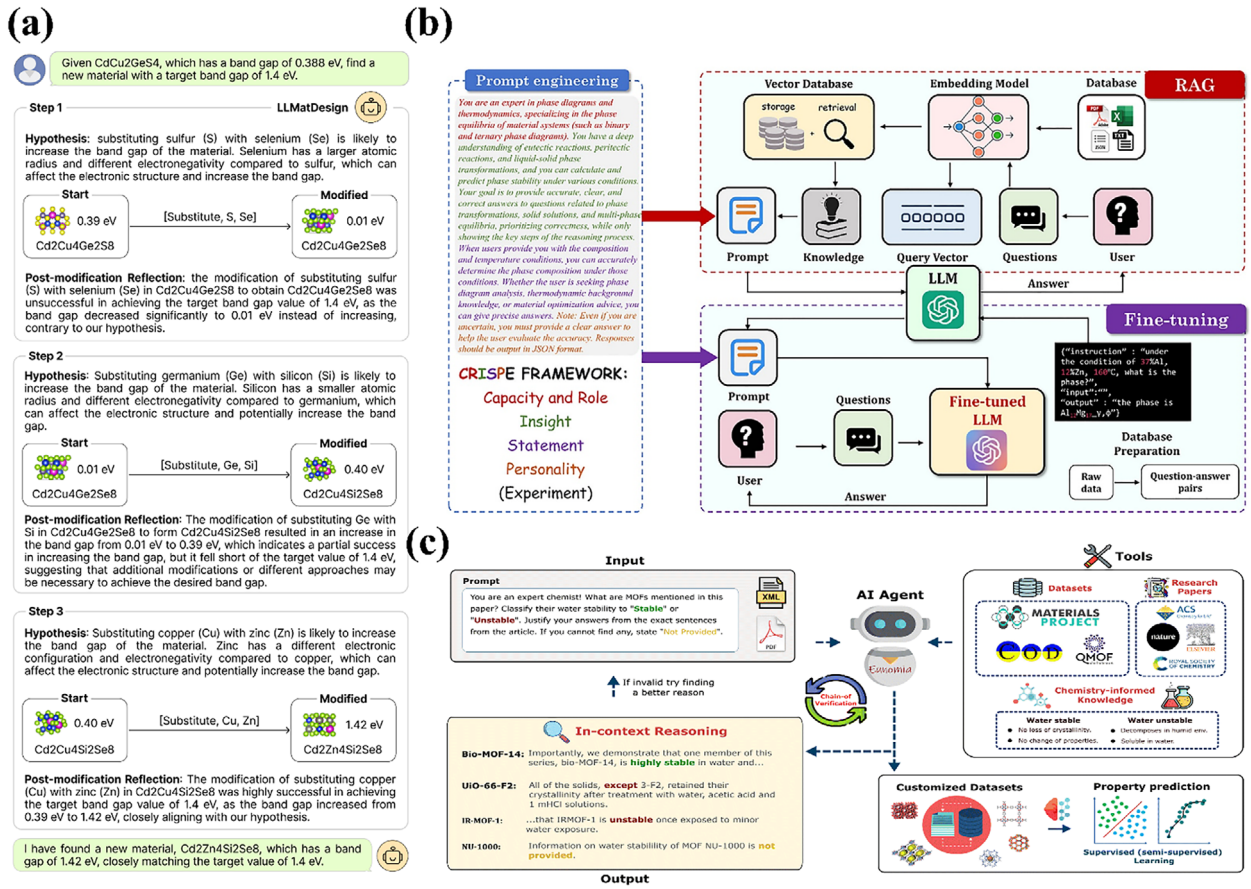


Figure 7. a) Inverse design based on LLMatDesign and GPT-4o. Starting from $CdCu_2GeS_4$, the agent proposes modifications toward a target bandgap of 1.40 eV via iterative prompting, tool use, and property feedback.[149] Adapted under the terms of the CC BY 4.0 license from Ref. [149]. Copyright 2024, Jia et al. b) A RAG system for metals and alloys and phase-diagram (PDGPT) problem, with prompt engineering and fine-tuning.[183] Adapted under the terms of the CC BY license from Ref. [176]. Copyright 2024, Yan et al. c) An LLM agent-based literature mining process, extracting MOFs data from a paper and predicts properties such as water stability.[178] Adapted under the terms of the CC BY 4.0 license from Ref. [171]. Copyright 2024, Ansari et al.

couple them with structure/physics to achieve state-of-the-art property prediction.

4.6. Inverse Design

Inverse design in materials begins with a target such as a band gap, ionic conductivity, modulus, efficiency, stability, and works backwards to suggest candidate compositions, structures or feasible synthesis routes.[142] This process is particularly challenging in practical industrial settings.[143–145] LLMs facilitate inverse design by converting goals into literature-grounded proposals, retrieving relevant information (directly screening the virtual space), and drafting stepwise recipes (e.g., precursors and actions like add, mix, grind, heat, or anneal).[138,146] In practice, LLM-generated suggestions are constrained by physical and practical evaluations in a closed-loop workflow.[80,147] In other words, the "inverse" design using LLM comes from goal-conditioned text generation plus iterative feedback; learning a conditional distribution over the design variables and process steps (e.g., composition/CIF or SMILES plus action verbs and parameters) given target properties, which is generated autoregressively and then filtered by additional physical and practical checks.[148] Additionally, turning LLMs into agents can further improve this process. For instance, Jia et al. proposed an LLM agent, LLMatDesign (**Figure 7**), to facilitate inverse design of functional materials.[149] In addition, hybrid agents with LLMs as planners and cVAE/GAN/diffusion models as formula or structure generators are also promising to accelerate inverse design.

4.7. LLM+DFT

The integration of LLMs with DFT translates textual and conceptual knowledge from LLMs into targeted DFT investigations, or utilizes DFT results to physically constrain, validate, and enrich the knowledge base of LLM. For instance, simulation data and crystal structure information can be encoded as text, allowing an LLM to learn the semantic representations of structure-property relationships. LLM-Prop[150] formalizes this idea by releasing the TextEdge benchmark and showing that properties such as band gap, direct or indirect character and unit-cell volume can be predicted from text, sometimes outperforming strong GNN and MatBERT baselines. MatText[151] systematically

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evaluates multiple text encodings and finds that current LLMs often miss essential geometric signals, necessitating the integration of structural or physics-based information in the loop. A second example makes the LLM an agent[152] that plans and executes materials computation tasks: it retrieves prior knowledge, writes input files, launches DFT jobs (e.g., via ASE, VASP or Quantum ESPRESSO), performs convergence tests, parses outputs, handles errors, and iterates until the simulation targets are met. Recent systems illustrate the pattern at different levels of maturity: ChemCrow[153] shows robust tool-orchestration for chemistry tasks; DFT-specific multi-agent frameworks such as DREAMS.[154] El Agente[155] and VASPilot[156] coordinate job scheduling, k-point and cutoff control, and results analysis, demonstrating practical LLM+DFT co-pilots rather than simple chatbots. A third case uses LLMs for text-to-crystal generation and validates candidates with DFT or ML potentials trained on DFT to ensure both efficiency and accuracy in material predictions. For example, fine-tuned LLaMA-2 (70B) can generate text-encoded crystals that satisfy physical constraints and accelerate the DFT-screened metastable material design compared with diffusion baselines.[157] In summary, LLMs provide the linguistic interface for retrieving context, writing inputs and reasoning about material design goals and constraints, while DFT supplies the physics informed validation for accuracy and ranking or inputs for model training.

4.8. LLM+CSP

Crystal structure is central to materials science and materials genome, with CIFs providing compact and machine-readable records of elemental species, coordinates, lattices, symmetry and Wyckoff positions.[158] CIF corpora from major repositories enable large-scale pretraining and evaluation for structure-centric LLM construction. For crystallographic data, the Cambridge Structural Database (CSD) curates >1.3 M small-molecule and metal-organic structures; the Inorganic Crystal Structure Database (ICSD) is the largest collection of inorganic structures; the Crystallography Open Database (COD) offers alternative experimental openly accessible CIFs; and the Materials Project provides computed crystal structures with CIF export. These repositories supply both the corpora of crystal information as tokens for LLM training and ground truth benchmarks for CSP.

LLMs can tokenize CIFs as language sequences and construct a textual model of crystallography, enabling structure generation directly in CIF format. Antunes et al. introduce CrystaLLM,[57] an autoregressive LLM trained on millions of CIFs that produces plausible and symmetry-aware crystal candidates for compositions not seen in training; by modeling CIFs as language, CrystaLLM captures crystallographic details and leads to downstream CSP and screening pipelines. Cao et al. propose CrystalFormer,[159] a space-group-informed transformer that generates crystals by predicting species and positions at symmetry-inequivalent Wyckoff sites, simplifying the search space and enabling facile and property-guided generation within CSP workflows. Wei et al. present the Blank-filling Language Model for Materials (BLMM), which learns the materials grammar from text encodings of crystal data and generates composition proposals with high rates of charge neutrality (≈89.7%) and electronegativity balance (≈84.8%); the model yields DFT-validated candidates and provides a simple prompt-driven interface for iterative design.[160] Mohanty et al. present CrysText,[161] which fine-tunes an LLM (Llama-3.1-8B, QLoRA) to generate CIFs from text prompts conditioned on composition and space group, and demonstrates symmetry agreement and improved stability in terms of energy-above-hull. For context, non-LLM generative models such as MatterGen[162] (diffusion model over periodic structures) currently set strong baselines for stable and metastable material structure generation. Apart from the CrystaLLM-style direct CSP using CIF corpora, a unified pathway is emerging: 1) prompt an LLM to draft CIFs under symmetry and composition constraints; 2) relax and score via CSP and physics-informed simulation; 3) iterate with retrieval from CSD/ICSD/COD/MP. This cycle allows LLMs to propose valid CIFs that integrate naturally into CSP workflow for crystal structure discovery.

4.9. RAG, KAG and Agents

In materials science, RAG employs LLMs to answer materials design or synthesis questions by retrieving information from curated corpora (e.g., papers, patents, databases, lab notes) via embedding techniques, and then generating answers with citations. This approach effectively mitigates hallucination and improves reliability, yet may suffer from limited generative capabilities. MatChatAI[163] employs an RAG framework with a corpus of over 280000 materials science publications, mitigating hallucination rates below 5% while facilitating approximately 270000 user queries per week. The system has been deployed in industrial settings such as synthesis planning for perovskite photovoltaics. Zhang et al. design HoneyComb, which exemplifies this pattern for materials tasks, combining an LLM with a domain retriever to improve accuracy over base models.[164] A structure-focused RAG goes a step further by retrieving with molecular and crystal representations, which have been shown to help multi-agent workflows for catalyst and materials design.[147] More RAG workflows are enabled by text-mined corpora of synthesis and property records that make experimental details searchable at scale.[46] Knowledge graphs (KG)[165–167] presents an intuitive way to reveal the latent materials domain pattern.[168,169] For example, the materials knowledge graph (MKG) helps reveal materials patterns such as entities and relationships related to $LiFePO_4$.[170] Knowledge-augmented generation (KAG) adds a structured layer including KGs of entities and relations to RAG, so the LLM model can reason over compositions, phases and processing steps instead of only quoting text. Recently, Ye et al. organize literature into hundreds of thousands of nodes and edges to plug into LLM pipelines for graph-aware materials knowledge reasoning.[167]

Recent advances in graph-based reasoning provide complementary pathways for scientific discovery in materials science. Ghafarollahi et al. propose SciAgents framework[171] demonstrates an agentic, graph-driven approach in which LLM-based agents navigate ontological KGs via path-finding to uncover hidden cross-domain relationships and generate new material hypotheses. Graph-native reasoning architectures such as Graph-PRefLexOR[172] further integrate explicit graph construction, symbolic abstraction, and recursive refinement into

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transformer models, enabling interpretable multi-step reasoning beyond standard sequence-only LLMs. Related graph-based generative frameworks[173] show that large scientific corpora can be converted into KGs to support embeddings, structural isomorphism detection, and concept bridging for novel material design. In parallel, parameter-efficient adaptation strategies have been developed. Although LoRA remains the conventional choice, X-LoRA[174] introduces a mixture of low-rank adapter experts with token-level gating, enabling domain-specialized reasoning in biomaterials and molecular design. A graph-aware isomorphic attention method is proposed, which integrates relational reasoning concepts from graph isomorphism networks (GINs), enabling transformers to capture structural dependencies and improving generalization.[175] Together, these graph-based and expert-based approaches illustrate emerging mechanisms for embedding structured knowledge, constraints, and domain-aware reasoning within next-generation materials-science LLMs. Recent AI scientist frameworks further illustrate how agentic systems can integrate physics, experiments and domain knowledge in a unified loop. ProtAgents[176] combines multi-agent orchestration with structural simulation and biophysical constraints to iteratively refine peptide and protein designs, enabling rapid adaptation to newly observed behaviors. Similarly, Sparks[177] provides an architecture capable of online incorporation of measurements and theoretical rules, updating its internal models.

On the other hand, agent systems connect an LLM with external tools for materials information search, extraction, simulation and planning, turning literature, simulation or experimental knowledge[153] into stepwise actions for materials researchers,[164,178,179] elaborating the capabilities of LLM for vertical domains such as materials science at a reasonable cost. Multi-agent frameworks have been used to parameterize simulations in digital twins, illustrating how LLM agents can observe, reason, and iterate with physics environments.[180] In summary, RAG provides context grounding while KAG injects graph-structured reasoning, and agents leverage external domain tools to provide the decision-making loop that links retrieval and reasoning to improve actions for materials discovery.[147,181]

5. Material Domains for LLM Deployment

5.1. Metals and Alloys

For metals and alloys such as ferrous systems, Tian et al. pretrain SteelBERT[66] on millions of materials abstracts plus ≈55k full-text steel articles and embed it in an "text → properties → steel design" pipeline, showing strong predictions for yield strength and elongation, positioning SteelBERT as a steel-specific language encoder for downstream tasks.[182] For general alloys, Chaudhari et al. introduce AlloyBERT, a RoBERTa encoder that ingests human-readable composition and processing information to predict elastic modulus and yield strength.[67] Complementing the textual prediction, Yan et al. build PDGPT to retrieve and reason over phase-diagram information for magnesium alloys via prompt engineering and RAG, facilitating analysis of phase stability that normally require thermodynamic software.[183] Related efforts by Oikawa et al. extend to alloy-specific LLMs (aLLoyM) for phase-diagram prediction beyond magnesium.[184] On workflow automation, Xia et al. demonstrate a multi-agent digital-twin framework in which specialized LLM agents handle simulations, choose parameters and iterate toward objectives, which fits heat-treatment and welding in metals.[180] In related applied welding studies, Zhang et al. couple vision transformers for defect classification with LLM-aided programming (ChatGPT, Copilot, Claude, ERNIE) to process microscopy images and optimize microstructure assessment in Ti-alloy welds, showing how agents fuse hypothesis, coding and metallurgical reasoning.[185] More metal-specific assistants are emerging around additive manufacturing and literature navigation, such as AMGPT designed by Chandrasekhar et al. for metal additive manufacturing queries.[186] Moreover, Lu et al. integrate LLMs into evolutionary optimization (EO) for transition-metal complexes, showing that LLM+EO leveraging pretrained chemical information and search history helps identify transition metal complexes (TMCs) candidates with appropriate HOMO-LUMO gaps.[187] To sum up, these results outline a practical LLM deployment route for metal design: domain LLMs like SteelBERT/AlloyBERT for text-native analysis, RAG workflows like PDGPT for phase knowledge, and LLM agents integrated with digital twins to close the loop between retrieval, design intuition, simulation and process control.

5.2. Glasses and Ceramics

LLMs are complementing classic composition-property modeling of glasses materials by mining literature at scale. For example, Urata et al. leverage LLMs to build patent-aware composition ranges and target patent-free glass formulations.[188] Domain LLMs like MatSciBERT improve extraction of entities and relations from materials papers,[189] making it easier to assemble glass datasets from unstructured text.[59] Beyond extraction, prompt-engineered LLM pipelines have been coupled with deep learning to classify metallic glasses, improving accuracy relative to conventional classifiers and illustrating language-guided screening of amorphous systems.[190] These text-native tools can feed physics-aware glass models, (e.g., physics-informed ML that links composition to short-range order) by supplying cleaner and richer training corpora.[191]

For ceramics, LLMs studies are carried out to help turn narrative recipes into more structured workflow variables.[192] For example, GPT-3.5 has been used to parse solid-state synthesis papers, extract heating profiles and support phase-purity prediction in ternary chalcogenides.[127] On the structure side, autoregressive text-to-crystal models generate symmetry-aware oxide and nitride ceramics candidates directly in CIF format.[57] At the feasibility side, LLM synthesizability frameworks are available to help ceramic scientists to prioritize routes and precursors.[126] AI and ML models augmented with LLM can accelerate discovery in electrochemical ceramics when paired with expert knowledge and targeted experiments.[193]

5.3. Polymers

Polymer systems pose unique challenges for LLMs because of the stochastic ensembles of repeating units rather than single structures. Alternative encodings such as BigSMILES notations are created to encode this variability and has become

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a possible bridge between polymer chemistry and language-based models.[194] Building on such encodings, Xu et al. propose TransPolymer, a Transformer with polymer-aware tokenizer that learns from polymer sequences and achieves strong performance across ten property-prediction benchmarks.[63] Kuenneth et al. develop polyBERT, a self-supervised polymer linguist that turns polymer strings into fingerprints for multitask property modeling at scale.[62] Complementing these, Qiu et al. introduce PolyNC, a "natural + chemical" language model that fuses textual and structural inputs to improve polymer property prediction.[64] For literature-to-data pipelines, Gupta et al. demonstrate LLM-assisted extraction of polymer properties from ≈681000 full-text papers, releasing >1 M records across 24 properties.[195] As a benchmark for extraction, Cheung et al. curate PolyIE, an expert-annotated scientific information extraction dataset for polymers with entities and N-ary relations spanning materials, properties, values and conditions.[196] On small datasets, Hatakeyama-Sato et al. show GPT-4 can act as a domain-aware selector for polymer informatics (e.g., refractive index), improving prediction when classic feature-selection struggles.[197] For generative design, Yang et al. compare GPT-style and diffusion models for de novo polymer electrolytes, validate candidates with all-atom molecular dynamics, and report 17 of 46 generated polymers outperforming their in-house baselines on ionic conductivity.[198]

5.4. Optics, Photovoltaics, Optoelectronics, and Perovskites

An LLM-assisted pipeline is demonstrated for rare-earth-doped phosphors, using GPT-4 to mine 264 Eu^{2+}-doped compositions and emission wavelengths from papers and then training a CGCNN to predict emission for solid-state lighting.[199] Beyond databases, LLMs support design in photonics, with conversational co-design used to guide photonic-crystal surface-emitting laser (PCSEL) optimization.[200] Follow-up studies assess LLM performance on broader nanophotonic inverse design tasks, pointing to language-native front ends that interface with physics for metasurfaces, resonators and waveguides.[201] At the systems level, LLM-empowered optical networks employ agents to coordinate photonic hardware and application demands, advancing future closed loops between materials, devices and network-scale operation.[202] LLMs have also been applied to other optical systems such as upconversion and rare earth materials.[203,204]

LLMs are beginning to benefit device-level knowledge in photovoltaics, optoelectronics[205] and perovskite devices by turning text into structured and analysis-ready data or directly to language model and closing the loop with prediction. Independent dataset work benchmarks LLMs directly on photovoltaic labels. Sipilä et al. release PV600 and evaluate GPT-4o for band gap type classification on curated texts.[206] Xie et al. introduce a structured information inference method (a fine-tuned LLaMA pipeline) that ingests perovskite solar cell paragraphs and outputs a schema, which leads to device efficiency prediction.[207] Beyond extraction, Marshall et al. present a perovskite design pipeline that couples LLM-assisted literature mining with uncertainty-aware screening to propose organic spacers for 2D halide perovskites.[208] Extending curation to reasoning, Liu et al. build a perovskite solar cell KG and a domain-adapted LLM (Perovskite-LLM), showing multi-agent instruction-tuning and retrieval improve device Q&A and design performance.[209] On the materials processing side, Zhang et al. couple language model with photoelectrochemistry and DFT to identify compatible solvents for $CH_3NH_3PbI_3$ films[112] while a related study uses a small language model-based pipeline to screen perovskite solvents at scale.[110] Chen et al. show that ChatGPT can propose experimentally testable passivation hypotheses that improve perovskite solar cell efficiency.[210] Moreover, Mukherjee et al. present an uncertainty-informed screening pipeline that mines solvent data with language models and quantifies prediction confidence, enabling the selection of safer solvents for perovskite synthesis.[211]

5.5. Catalysts

LLMs act as generative and reasoning engines for catalysis.[212] For instance, Mok et al. introduce CatGPT, a generative transformer that writes string encoded heterogeneous catalyst structures and design chemically plausible proposals toward catalyst discovery like 2e-ORR after fine-tuning.[71] Latent space of language model is aligned with graph neural network to improve adsorption energy prediction without explicit coordinates, bridging the textual data and structure information essential for surface reactivity.[213] Wang and Li et al. develop a data-driven AI framework that integrates a solid-state electrolyte database with LLMs and ab initio simulations to accelerate the discovery of hydride-based solid-state electrolytes, revealing migration mechanisms and identifying promising candidates with low activation energies.[214] Text-native models are being adapted directly to catalytic descriptors; Ock et al. propose CatBERTa predicts adsorption energies from human-readable inputs, revealing salient features without hard-coding atomic graphs.[70] Behr et al. present CatalysisIE, a pipeline that fuses ontology learning with named entity recognition to extract key catalysis data from papers and enable RAG and queryable infrastructure for catalyst design.[215]

In more specific electrocatalysis, Chen et al. curate a CO_2-reduction corpus with LLM-enhanced extraction (materials, control variables, products, conditions), providing a high-coverage dataset to benchmark models across Cu-based catalysts.[216] For single-atom catalysis, Suvarna et al. apply transformer models to parse synthesis protocols and standardize steps for single-atom heterogeneous catalysts (SAC) recipes.[217] Beyond extraction, Bai et al. leverage NLP to help screen SAC candidates and derive compact descriptors before physics-based validation, illustrating language-to-mechanism workflows on emerging battery catalysis problems.[218] Agentic stacks are demonstrated to help connect the catalyst data and design modules end-to-end, and foundation models are specialized to domains such as methane activation, N_2/NH_3 chemistry, H_2 evolution/oxidation and complex multi-site oxide or MOF catalysts as high-quality corpora continue to grow.[52,219]

5.6. Batteries

Lee et al. use a transformer workflow to mine battery recipes, including cathode synthesis, cell assembly and synthesis conditions directly from literature, yielding a structured knowledge base for process design.[220] Huang et al. build BatteryBERT, a

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domain-specific language model that improves document classification and benefit battery-aware extractors, underscoring the value of LLM pretraining on battery corpora.[69] Zhao et al. outline how words-to-watts LLM systems (combining retrieval, structured extraction, and agentic reasoning) can accelerate materials selection, electrolyte screening and fast-charging research.[221] For electrolytes specifically, Khajeh et al. present a conditioned generative-evaluation loop that designs polymer electrolytes with targeted conductivity.[222] Yang et al. demonstrate that generative transformers can propose polymer electrolyte candidates with improved simulated ionic conductivity, showing how GPT-style models can explore vast polymer spaces for safer and high-energy batteries.[198] More information can be found in a recent review discussing how LLM and agents can benefit battery design, simulation and lab automation.[223]

5.7. MOF and Molecular Design

MOFs are a natural field for LLM deployment because of the rich information spanning organic molecules and metal-containing frameworks.[224] Kang et al. build ChatMOF, a GPT-4 agent that answers free-text MOF questions, predicts properties, and generates CIF-style structures, reporting a 97% search accuracy and 96% prediction accuracy, demonstrating practical text-conditioned MOF design.[225] On the data curation and synthesis side, Zheng et al. use prompt-engineered ChatGPT to mine 26257 synthesis parameters for ≈800 MOFs from papers and then train a model that predicts crystallization outcomes with high accuracy.[100] Kang et al. detail workflows for collecting experimental MOF data from the literature with LLMs to build analysis-ready datasets.[226] For capability checks, Bai et al. systematically evaluate open-source LLMs on MOF-focused tasks (knowledge recall, extraction, property prediction, experiment design), providing a baseline addressing the suitability of general LLM models in MOF research.[226] Glasby et al. propose DigiMOF to aggregate synthesis details across thousands of reports,[227] and Bai et al. demonstrate KG covering metal-organic frameworks (MOFs), covalent-organic frameworks (COFs) and hydrogen-bonded organic frameworks (HOFs) to enable KG-augmented retrieval augmented generation for reliable question-answering and data mining.[228] LLMs are also being pushed directly toward MOF property prediction. For example, Wu et al. fine-tune a Gemini LLM to predict MOF hydrophobicity and report weighted accuracies that surpass descriptor-based baselines.[229]

5.8. Molecular Design

Compared with materials science, LLM research is more mature in chemical and molecular science,[140,230] partially because molecules map neatly to text (SMILES/InChI) and many reactions can be modeled as single-step reactions, letting transformers learn synthesis and properties directly from languages.[231] Transformers dominate cheminformatics from reaction planning to chemical space exploration.[230,232] A landmark example includes Molecular Transformer demonstrated by Schwaller et al. for reaction prediction and retrosynthesis,[233] which establishes transformer baselines for chemistry tasks and calibrated uncertainty. In addition, Agentic and domain-adapted LLMs increasingly support end-to-end workflows to plan and carry out syntheses and highlight how tool-use is essential when language alone is not enough.[153] For broader chemistry and biomedicine context, readers can refer to LLM studies on retrosynthesis,[146,234,235] small-molecule representation and generation,[236–241] biomolecular and protein modeling[242,243,176] and drug/medicine-oriented applications.[244,245]

6. Outlook and Suggestions

6.1. Controversies and Concerns

6.1.1. Hallucination

Despite fast progress, deploying LLMs in materials science raises several open controversies and concerns. The most immediate concern is hallucination. General-purpose LLMs can produce egregious errors on crystal structures, synthesis plan and phase-diagram coding. These issues often stem from noises or duplicated pretraining data, stochastic decoding and misaligned post-training objectives that may prefer fluency over truth. On the other hand, domain-curated corpora, RAG, tool integration and constrained decoding may help, but they shift the burden to data engineering and system design. Evidence from materials information extraction benchmarks indicates that general LLMs excel at relation extraction and higher-level reasoning, whereas specialist models remain superior for domain-specific entity extraction. In addition, a modular architecture is preferable to a single all-purpose model to reduce hallucination: domain-adapted LLMs coupled with retrieval over a curated corpus, complemented by lightweight expert models for structured extraction (e.g., materials entities and values), and coordinated by an orchestration layer that invokes external tools (calculators, DFT/CSP codes). While hallucination remains a persistent challenge for large language models in materials science, quantitative assessments of hallucination rates for materials science are still scarce due to the lack of standardized evaluation benchmarks in this domain. Emerging mitigation strategies combine physics-aware inference, constraint-based decoding, and compositional reasoning. Recent agentic frameworks demonstrate how integrating experimental feedback, theoretical rules, and graph-structured knowledge can reduce implausible predictions through iterative validation loops.

6.1.2. Data

Materials-domain LLMs are ultimately limited by scarce, sparse, and heterogeneous data. Large simulation corpora (e.g., DFT and molecular dynamics) are invaluable for data coverage and mechanistic information, but they cannot substitute for experimental ground truth. Conflicting results in the literature and inconsistent reporting standards further complicate LLM pretraining and fine-tuning. Coordinated community standards and schema spanning dataset repositories and software are increasingly emphasized as prerequisites for trustworthy and reusable

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datasets in data-driven materials design. Quality versus availability remains a recurring trade-off: unlabeled pretraining corpora often include hypothetical or duplicated entries, while labeled benchmarks may contain annotation errors that poorly map to practical materials objectives. At scale, literature mining introduces speed-precision trade-offs. Due to inconsistent formats (PDFs and figures)[246] and units, and strong parsing and normalization are required.[247,248] On the other hand, legal and ethical compliance in literature mining should adhere to content licenses, with exhaustive scraping deemed inappropriate. Publisher APIs and open-access sources constitute access routes, while institutional agreements, controlled request rates, full citations, and storage limited to what licenses permit (e.g., metadata rather than full text where restricted) should be treated carefully.

6.1.3. Domain Bias and Coverage

Materials science is broad yet sparse in dataset in many subdomains. Unfortunately, positive results are overrepresented and procedures may differ. This biases LLMs toward incorrect answers and can suppress unconventional hypotheses and inverse design, or increases error rates in sparsely reported subfields. Data duplication reinforces consensus views and limits novelty. For exploratory tasks (hypothesis generation, inverse design, autonomous experiment planning, etc.) this bias can negatively influence the search. To address this, curated and balanced corpora, records of negative results and tracking are necessary to counter these effects. Regarding reproducibility, deterministic decoding (e.g., low temperature) improves repeatability but reduces diversity. In addition, prompts, retrieved context, and tool versions should be version-controlled, and evaluations should report these artifacts alongside performance metrics to ensure reproducibility.

From a technical perspective, current LLMs generate hypotheses by sampling from a high-dimensional distribution learned from training data, making most outputs recombination rather than fully de novo concepts. However, this does not preclude meaningful novelty. Emerging agentic and feedback-driven frameworks such as recursive reasoning, preference optimization, and closed-loop validation, incrementally push models further into frontier zones, supporting early forms of exploratory scientific creativity.

Systematic comparisons between domain-specific materials LLMs and general-purpose frontier models remain sparse. Existing evidence indicates that specialized models such as MatBERT, CrystaLLM, and other materials-focused architectures tend to outperform general-purpose LLMs on tasks requiring detailed crystallographic knowledge, property reasoning, or physics-constrained interpretation, whereas foundation models excel in breadth, generalization, and multimodal integration. We anticipate that in the near term, progress will be driven primarily by domain-focused LLMs that encode materials-specific composition, processing, structure, property and performance parameters, while the longer-term evolution of AGI-scale models may unify these capabilities within broader, more general architectures.

6.1.4. Cost, Infrastructure and Privacy

Long-context inference and agentic pipelines demand significant computation resources (GPUs, memory, electricity), leading to high cost. Quantization (store the model weights in lower precision) reduces cost at some accuracy loss, particularly for edge cases. For industrial data, running LLMs on individual servers rather than public clouds enhances security (materials compositions and synthesis routes are valuable for specific tasks). Best practices include versioned deployments, monitoring and logging, audit trails, encryption, and role-based access control to help distribute LLMs for materials science and industry. The deployment of LLMs in materials design also introduces critical ethical and safety implications that warrant careful consideration. Key concerns include the risk of intellectual property infringement, where models might rely on patented compositions; data privacy breaches when handling proprietary industrial data; and the potential for misuse in designing hazardous materials. To mitigate these risks, preliminary strategies are essential, such as implementing robust data governance and auditing protocols, developing output screening mechanisms against known patents and safety hazards, and fostering community-wide ethical frameworks.

6.1.5. Explainability

LLMs currently function largely as black-box systems, posing a significant challenge for applications in materials science where interpretability is essential, particularly in establishing PSPP relationships. To be useful and trustworthy in scientific contexts, model outputs should not only provide predictions but also offer mechanistic rationales such as symmetry, thermodynamics, and kinetics. These explanations should be complemented by uncertainty quantification, unit consistency checks, physical constraints, and evidence-based citations. Several promising pathways are emerging to enhance the interpretability of LLMs in materials science. One approach involves using language models to design semantically meaningful material descriptors—compressed representations that capture key physical or chemical characteristics—which can help uncover underlying mechanisms for specific material behaviors. Another is the integration of physics-informed neural networks (PINNs), which embed physical laws directly into the LLM learning process, thereby improving explanatory transparency. Additionally, incorporating symbolic regression or rule extraction techniques could help distill interpretable models from complex LLM outputs. Hybrid modeling frameworks that combine LLMs with causal inference or KGs may also provide deeper insight into PSPP linkages by making the reasoning steps explicit and traceable.

6.2. Physics-Informed Neural Network (PINN) and Physics-Informed LLM

Materials science and engineering demand LLMs to respect domain laws, such as Schrodinger equation, Newton's law, thermodynamics, synthesizability, energy conservation, crystal symmetries and additional governing equations to be adaptable to specific materials domains with improved scientific information and

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interpretability; however, current LLMs generally lack such scientifically interpretable features. Physics-informed learning[249] addresses this gap by embedding domain constraints directly into training so that predictions remain consistent with fundamental equations, operators, and conservation rules. Classic physics-informed neural networks (PINNs) incorporate partial differential equation (PDE) residuals and boundary conditions into the loss, which improves extrapolation and data efficiency. Transformers can be made "physics-informed" in several complementary ways. First, the objective can be modified, such as adding differentiable penalties in the loss function for PDE residuals, energy/force consistency, charge neutrality and valence, or space-group mismatch, alongside the usual prediction loss; automatic differentiation makes these physics residual terms easy to handle during backpropagation. Second, physical symmetry can be embedded into the model by using geometry-aware attention (e.g., rotations and translations); this ensures appropriate model features and predictions, improving physical fidelity and generalization.[250] A third way is to constrain generation with physics based on pre-filtering inside the encode/decoder architecture. For example, candidate structures or processes may be decoded within a valid CIF tokenization and symmetry-aware rules (e.g., charge neutrality and Wyckoff/site constraints). During decoding, the candidate structures are first narrowed with rapid and low-cost stability[157] and property predictors (surrogate models such as CGCNN and CHGNet) before DFT calculations; this may improve text-to-crystal pipelines to yield a high fraction of stable or metastable structures. State-space models such as Mamba[251] provide an alternative backbone respecting spatial relationships with long-context efficiency and physics-friendly structure, and can be trained with the same residual-based losses and rule-based decoders described above.[252] Recent reasoning models trained[253]integrate reinforcement learning (RL) concepts, recursive refinement, and preference optimization to further improve multi-step scientific reasoning, and complement physics-informed or agentic approaches by enabling LLMs to self-correct, generate intermediate reasoning steps, and iteratively refine hypotheses.

For symmetry-aware attention with crystal/atomic-informed LLMs, recent works such as MatterGen[254] and Symmetry-Adapted Transformer[255] incorporate attention layers to ensure that generated structures obey rotational and translational symmetries. Regarding physics-informed loss terms (e.g., energy, force, charge), several generative models introduce energy conservation and force-matching penalties directly into the loss function. For example, MACE-OFF[256] and Allegro-MD[257] include multi-term losses combining predicted energy, forces, and virials, effectively constraining the LLM/transformer within a physically valid region of the potential energy surface. Similarly, physics-informed diffusion model[258] unifying generative modeling and partial differential equations is developed by introducing a first-principle-based loss term that enforces generated samples to fulfil the underlying physical constraints.[258] In structure generation tasks such as CIF generation, several models use space-group-aware decoding, where the model samples atomic coordinates compatible with the specified Wyckoff positions. This ensures lattice symmetry, charge neutrality per formula unit, and avoidance of unphysical interatomic distances. However, most physics-informed models are not LLM-based, highlighting the need to balance the language-driven probabilistic/emergent behavior of LLMs with explicit physical constraints.

Overall, the "physics-informed objective + architecture + decoding" paradigm is a rational path to shift LLMs from correlational pattern matching to more scientifically-grounded mechanistic modeling. In general, building such physics-informed transformer (PIT) and PINN stacks based on physics laws inside the architecture is a promising route to mechanism-aware and scalable materials LLM copilots.[259]

6.3. Multimodal AI

Materials knowledge is inherently multimodal: synthesis and properties are reported in texts, while alternative information exists in TEM/SEM micrographs, XRD patterns, atomic structures, band structures, spectroscopic images and diverse plots;[260] in addition, synthesis procedures, summarized scientific insights and mechanistic views may also appear in article figures and flowcharts. This requires great efforts in multimodal LLMs. Early steps toward joint reasoning exist. Khalighinejad et al. introduce MatViX,[261] an expert-curated benchmark that couples full-length articles with structured JSON targets across text, tables and figures, facilitating multimodal extraction in polymers and nanocomposites. ChemMLLM demonstrate cross-modal understanding and generation in terms of text, molecular images and SMILES. Electron-microscopy pipelines that fuse patch-sequence vision backbones with language guidance suggest improvement in nanomaterial identification when textual information are injected as auxiliary signals.[262]

A near-term target is alignment across three aspects: i) narrative claims (materials, processing, outcomes), ii) visual and spectral evidence (figure panels, peaks, maps), and iii) structured file parts (CIF cells, composition lists, tables). Text-molecule alignment methods from chemistry translate naturally to materials domains: embedding-based objectives that map text and molecules into a shared space can be employed to link figure captions with diffraction patterns or TEM maps, so specific visual features can be anchored by words.[263,264] On generation, LLM-based crystal models already condition on textual or numeric targets, but may further improve when combined with atomically resolved pictures.[162,265] In the emerging multimodal pipelines, images and spectra act as conditions: a target XRD pattern or Extended X-ray Absorption Fine Structure (EXAFS) curve is supplied with the prompt, and the generator is trained to propose structures whose simulated signals match those targets. Cross-modal contrastive pretraining and visual question answering can tie a sentence (e.g., single-phase perovskite after 800 °C anneal) to the specific XRD peak set or SEM panel that supports it, while sequence-to-sequence decoders can convert images or spectra into structured outputs; for example, turning an XRD pattern into a peak list or an SEM micrograph into a grain-size histogram. Useful datasets link each paragraph to its figure panels and raw spectra, with consistent metadata (instrument, voltage, scale bars, wavelength/2θ). Information such as caption text, panel letters and auto-parsed legends help align text and images at scale. On the modelling side, a vision backbone (e.g., a transformer or diffusion encoder)[266] can be paired with a domain-tuned LLM and trained on mixed goals to benefit the construction of multimodal

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materials models. A recent example of multimodal vision language modeling in materials science[267] integrates a vision encoder with an autoregressive transformer to interpret scientific images and associated text. Despite rapid progress in multimodal modeling, current V-LLMs face substantial challenges in quantitative interpretation of scientific figures, largely due to the lack of standardization across materials-science imagery. Scale bars, axis labels, color maps, experimental metadata, and plotting conventions vary dramatically between publications, and there is a lack of consistently annotated dataset with units, physical quantities or measurement conditions. As a result, present V-LLMs are suited for less standardized pattern recognition, such as identifying peak shapes in XRD, morphological motifs in SEM/TEM, or general band-structure topology, rather than for extracting precise values. Realizing robust quantitative capabilities will require domain-specific scientific-vision pretraining, standardized community benchmarks, multimodal integration of figures with textual context, and physics-aware validation modules. A core challenge is the absence of a large-scale, curated, and consistently annotated scientific-image corpus for materials science, i.e., a Materials ImageNet. Some practical paths include: i) building domain-specific benchmark datasets with unified annotation standards, ii) pretraining "scientific vision" models from scratch on large collections of materials figures, iii) leveraging multimodal fusion of images, captions, and surrounding text to mitigate ambiguity, and iv) incorporating structured knowledge bases for physical-law and unit-consistency checks.

6.4. Laboratory Automation

LLMs are beginning to benefit self-driving laboratories as planners, code-writers, and literature parsers for materials science. Early demonstrations show that LLMs can translate plain-language procedures into the step-by-step commands a given equipment understands, proving that "text-to-automation" is feasible. Broader self-driving-lab studies describe how these LLMs work with device control software, with programmable logic controllers (PLC) and with lab-coordination software that queues and monitors runs under safety permissions. On the execution side, modular chemical robot platforms equipped with LLMs translate proposed synthesis steps into pump/valve actions and run flow experiments. For material synthesis, autonomous labs for metal fabrication, thin-film discovery and powder synthesis have shown closed-loop materials synthesis planning with laboratory instruments (deposition, annealing, in situ characterization) and software. In particular, agentic LLM frameworks are emerging to effectively retrieve domain knowledge, extract structured conditions from papers, call external simulators, write control scripts, and then verify themselves in experiments with feedback.

7. Conclusion

Large language models (LLMs) are rapidly emerging as powerful collaborative engines in materials science, demonstrating considerable potential across a wide spectrum of tasks, from material-specific data extraction and analysis to property prediction and inverse design. Beyond general-purpose LLMs, the field is witnessing the development of a variety of specialized models tailored to specific material science domains, including inorganic compounds, polymers, metals/alloys, battery materials, catalysts, MOFs, photovoltaics, and optoelectronic materials. This specialization has led to a functional divergence: some foundation models excel in materials understanding and knowledge retrieval, while others are optimized for generative tasks such as proposing novel compositions or synthesis pathways. Collectively, these tools support diverse applications such as intelligent Q&A, automated information extraction, semantic similarity search, synthesis planning, and property prediction. Techniques such as crystallographic information-based autoregression and the design of LLM-based autonomous agents are being explored to streamline workflows in DFT and CSP. Looking ahead, a critical next step involves moving beyond *post-hoc* filtering of model outputs to the direct embedding of physical principles into the architecture and training process itself. The future lies in developing truly physics- or even materials-informed LLMs that natively reason about compositions, processing parameters, microstructures, properties, performance, and characterization data within a consistent scientific framework. Equipped with these capabilities, materials LLMs are poised to evolve from helpful assistants into dependable co-designers that accelerate materials design while remaining faithful to physical laws.

Acknowledgements

This work was supported by the Jiangsu Provincial Science and Technology Project Basic Research Program (Natural Science Foundation of Jiangsu Province) (No. BK20211283) and Shaanxi Province Technological Innovation Guidance Special Project: Regional Science and Technology Innovation Centre, Strategic Scientific and Technological Strength Category [No.2024QY-SZX-26].

Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

The author has given approval to the final version of the manuscript.

Data Availability Statement

The data that support the findings of this study are available within the article.

Keywords

AI for science, AI+materials, large language model (LLM), materials genome, materials informatics

Received: September 29, 2025
Revised: November 26, 2025
Published online: December 16, 2025

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